

TBDP DATASET ACCESS GUIDE

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Revision log:

Version 1.0	10/1/2022	Shradha Chaphekar	Initial version
Version 1.1	2/7/2023	Shradha Chaphekar	Added queryTimeout configuration
Version 1.2	6/16/2023	Anil Gadi	Added ODBC SCCM installation
Version 1.3	6/21/2023	Anil Gadi	Removed the client secret keys
Version 1.4	6/22/2023	Anil Gadi	Updated instructions to retrieve keys from AWS Secrets Manager
Version 1.5	6/28/2023	Anil Gadi	Updated instructions to update the secret key in tableau tdc file

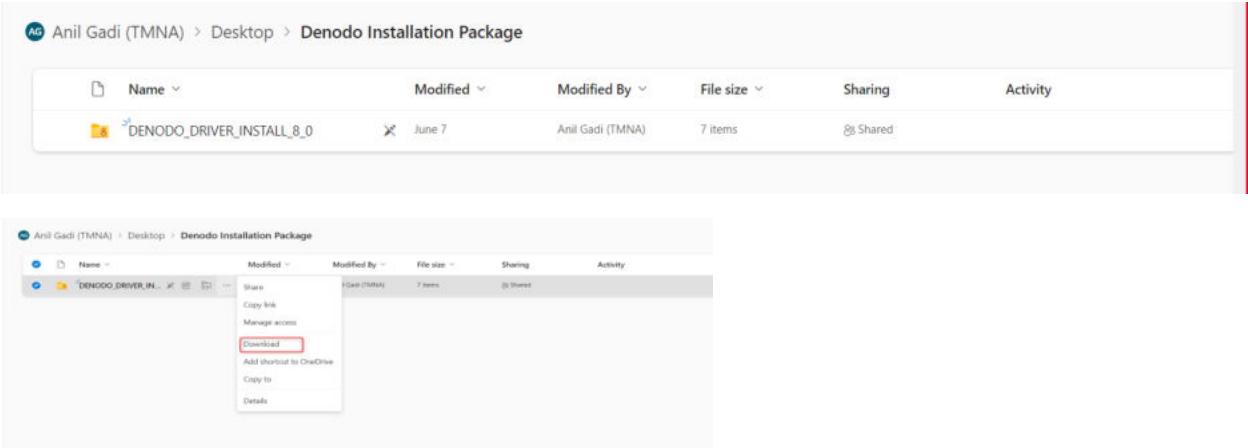
NOTE: Need Local Admin access to install ODBC Driver and required tools or reach out to helpdesk to install the driver.

1. Install Denodo Driver and Creating the DSN for Denodo

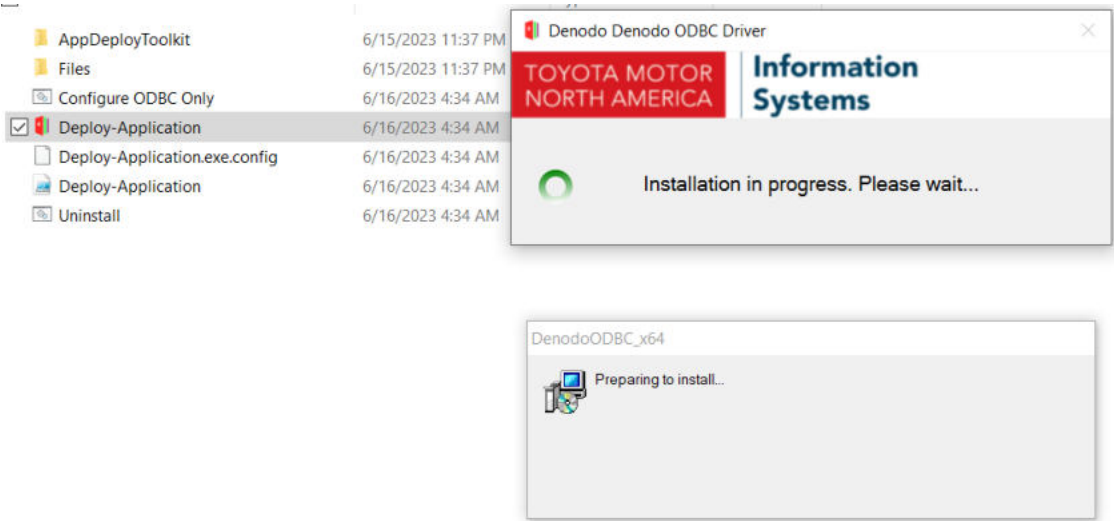
Please follow the below process to install the Denodo ODBC Driver and setup the ODBC connection.

Step1: Please click on the below link and **download** the Denodo Driver installation folder on to your desktop local and unzip it.

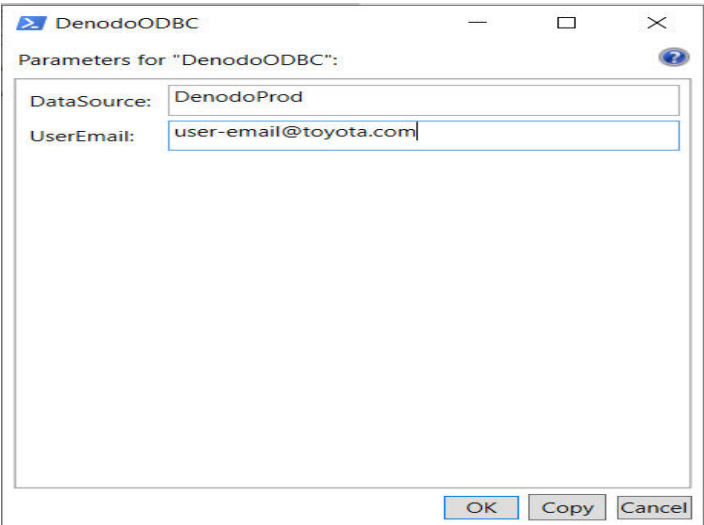
https://mydocs.toyota.com/:f:/g/personal/anil_gadi_toyota_com/Ep78fOROCzZBhfHLKSvTpFUBrNkDr49-VteTrl4hjMsA9g?e=QCeuPU



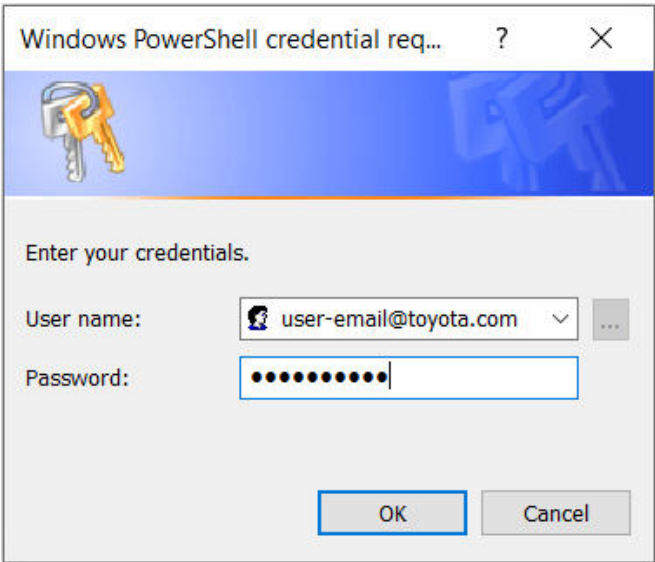
Step2: Double-click the below highlighted file (Deploy-Application) to get the installation process started.



Step3: Please enter Toyota email and Datasource name as ‘DenodoProd’ as shown below.

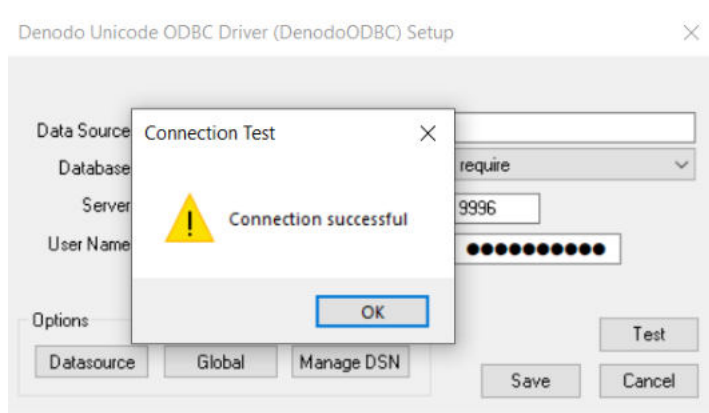
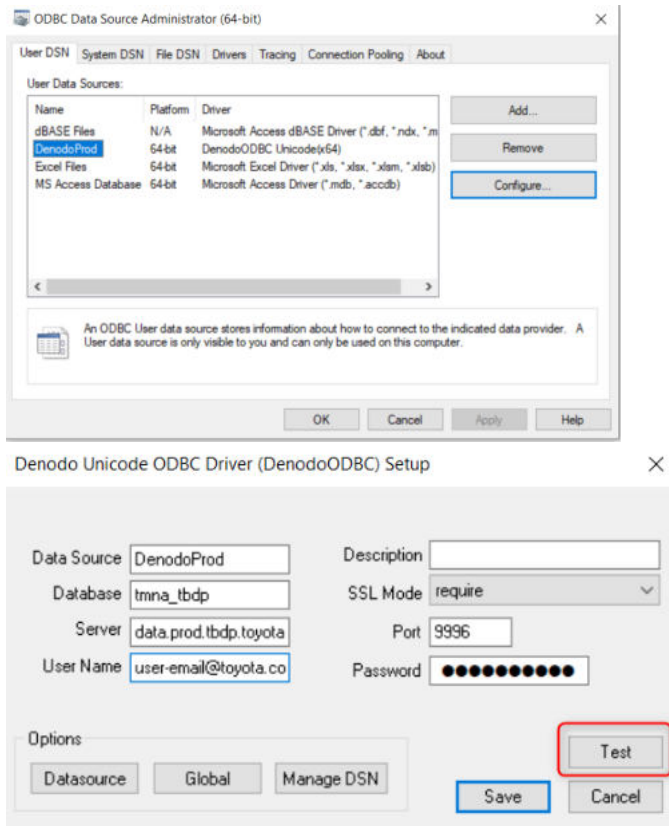


Step4: Please enter the Toyota network password and click ok. Installation will be successful as shown below



Step5: Open the ODBC admin tool and validate if the Denodo ODBC connection is successful as shown below.

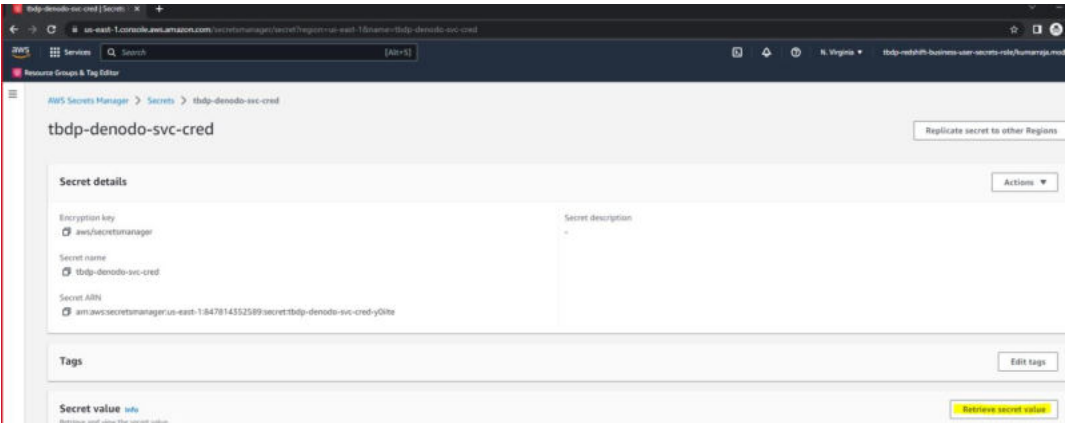
Go to Control Panel > Administrative Tools > Data Sources (ODBC) or Search for ODBC Data Sources (64-bit)



Step6: Click on Datasource, click on Page3 and configure the client secret value as shown below .

<https://login.microsoftonline.com/8c642d1d-d709-47b0-ab10-080af10798fb/saml2?SAMLRequest=jZFN8MwDIb%2FSvA9iZO6iWOSQPofhQ5G93HYTUnVNVDbneVOY79%2BaUtPg7Gr0Cs9j1QS6ONJNYM%2FmC1%2BDEg%2B%2BNJHQxUbnFEWqCdIQCMp36mn5mGjOogrjR524IEF60XFVIlspJRNm5kJMU9mUohJPm3EUqYJXxRLFryio96aio3hMUM04NqQB%2BPHEk95mPAwTZ%2FTRCVcJXkkc%2FF26XsEov6MFdvDkZAFDRE6P06aW0ODRveE7tx3%2BLLdVOzg%2FYIUHFp%2FbnoTwSdFoOHbmqizOr6lspubukr%2FbXhy1tvOjpGVdR1eD3THqMurgfvPpeBOzOoL34g3csVZUbQ4RQyLLBOhmiAM292uC9s84TDd59ke2zK%2BbanL%2BPeT6h8%3D&RelayState=https%3A%2F%2Fconsole.aws.amazon.com%2Fsecretsmanager%2Fhome%3Fregion%3Dus-east-1%23%21%2Fsecret%3Fname%3Dtbdp-denodo-svc-cred>

Select Role : **tbdp-redshift-business-user-secrets-role**

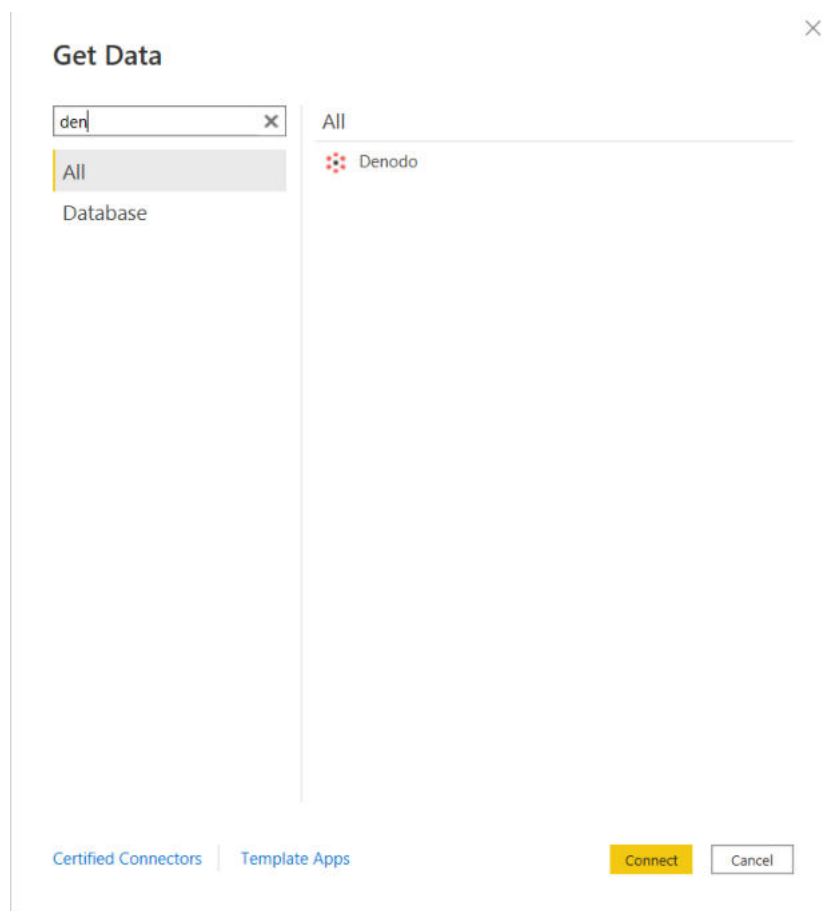


Click on Retrieve Secret Value to get clientID and clientSecret and update the values.

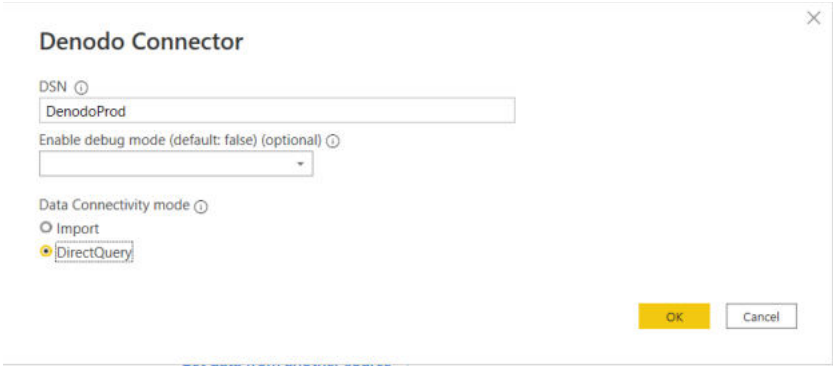
2. Connecting Power BI to Denodo

1. Power BI and Denodo ODBC integration

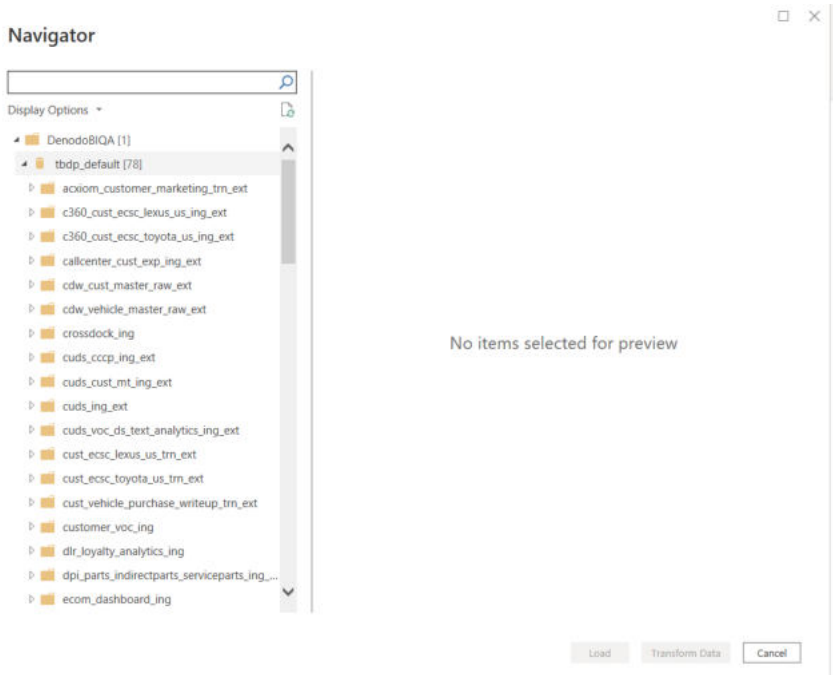
- a. Open Power BI desktop
- b. Click on Get data and click on more and search for “Denodo”



- c. Provide the DSN name that is created in Step 2 and choose direct query option. Click ok



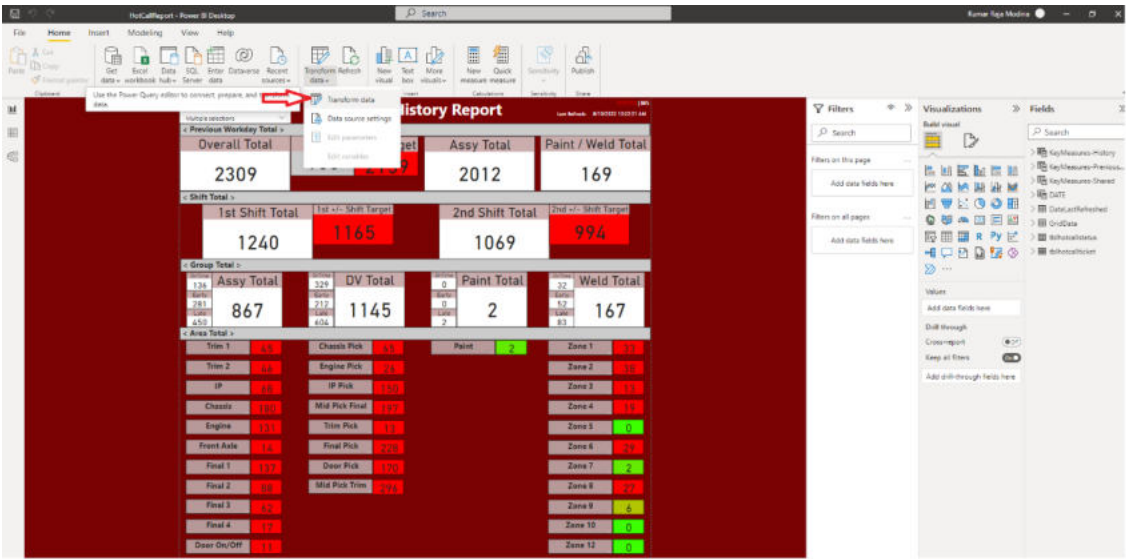
d. Once connected it will show list of schemas



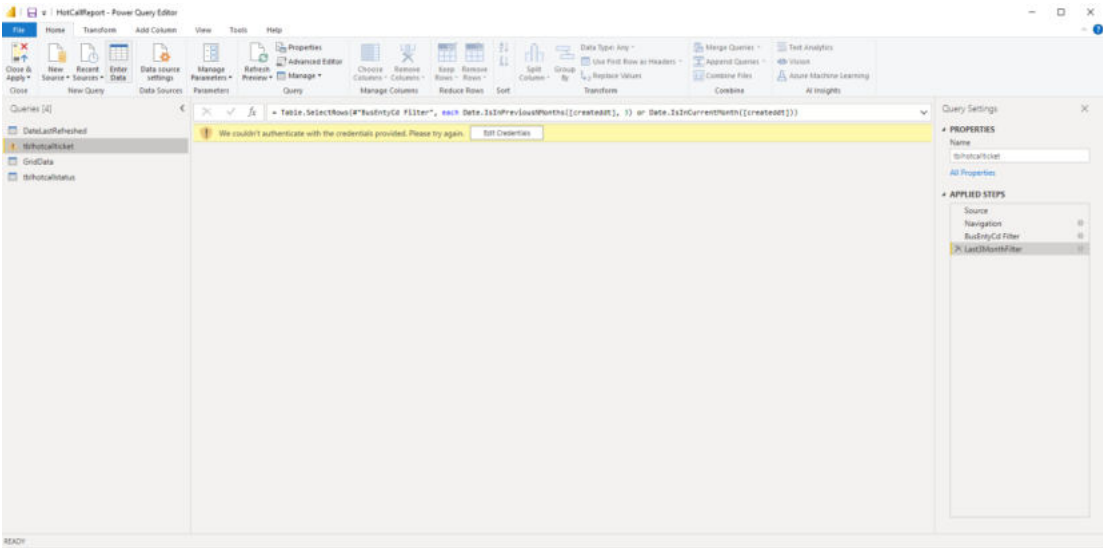
e. Select the table/s from the schema and click on load to derive insights

2. Rewire the current Power BI Report pointing it to Denodo

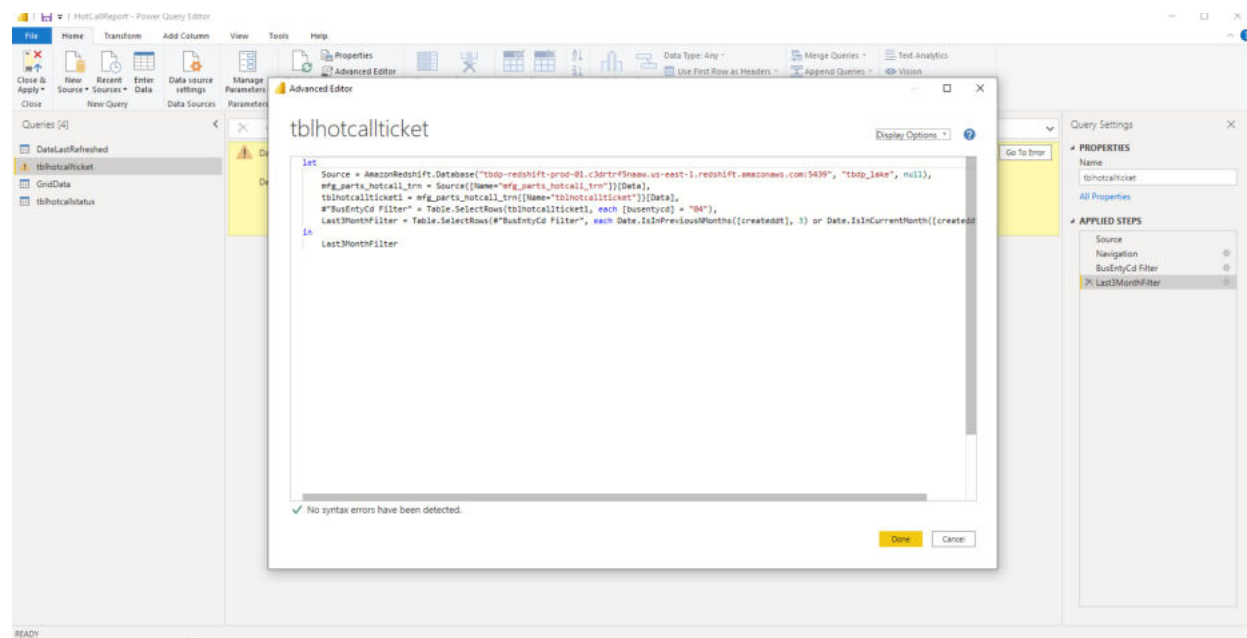
- a. Open Existig Power BI Report
- b. Go to Transform data > Transform data (Refer below screenshot)



c. Select Redshift table and Click on Data source settings (Refer below screenshot)



d. Click on Advanced Editor (Refer below screenshot)



- e. Replace Redshift connection string with Denodo connection string as follows.

Existing :

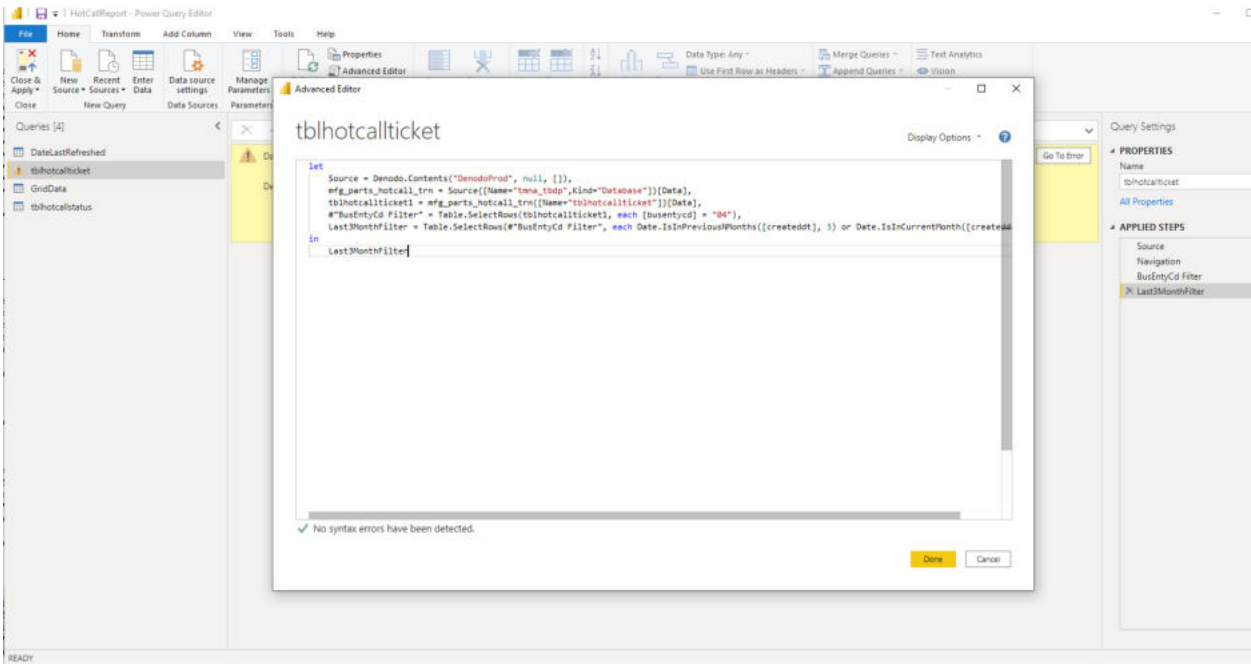
let

```
Source = AmazonRedshift.Database("tbdp-redshift-prod-01.c3drtrf5naaw.us-east-1.redshift.amazonaws.com:5439", "tbdp_lake", null),
mfg_parts_hotcall_trn = Source{[Name="mfg_parts_hotcall_trn"]}[Data],
tblhotcallticket1 = mfg_parts_hotcall_trn{[Name="tblhotcallticket"]}[Data],
#"BusEntryCd Filter" = Table.SelectRows(tblhotcallticket1, each [busentrycd] = "04"),
Last3MonthFilter = Table.SelectRows(#"BusEntryCd Filter", each
Date.IsInPreviousNMonths([createddt], 3) or Date.IsInCurrentMonth([createddt]))
in
    Last3MonthFilter
```

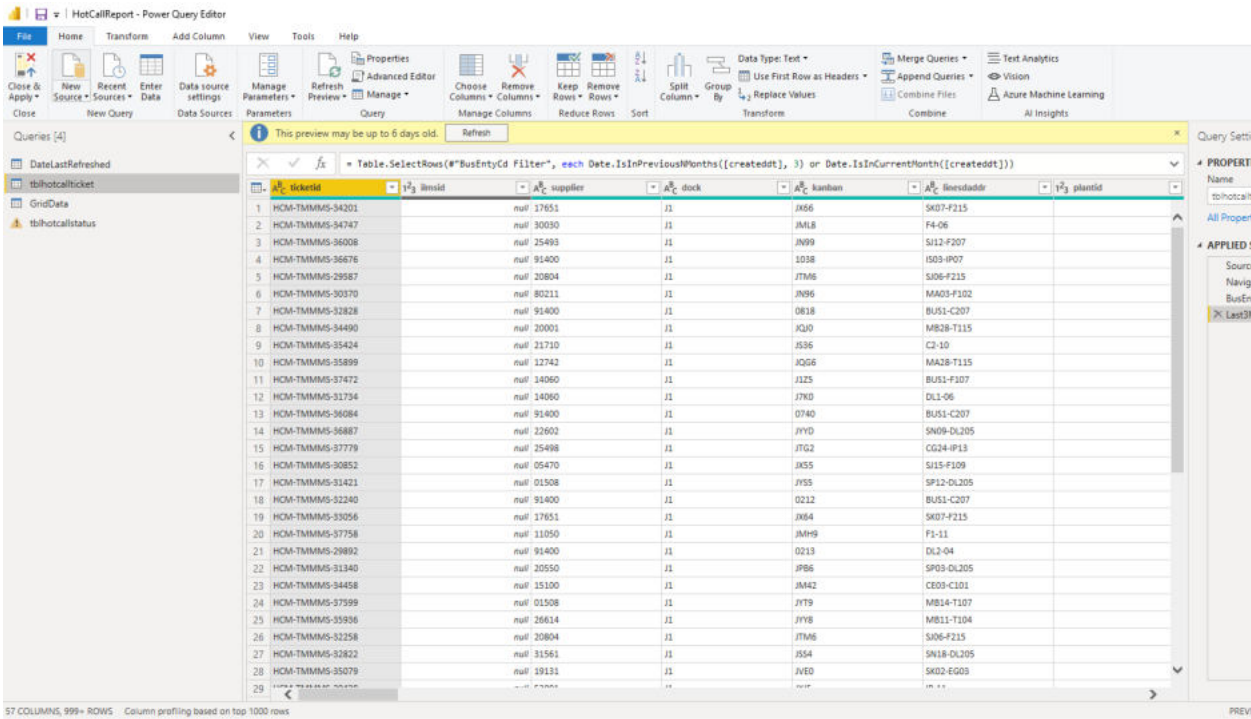
New:

let

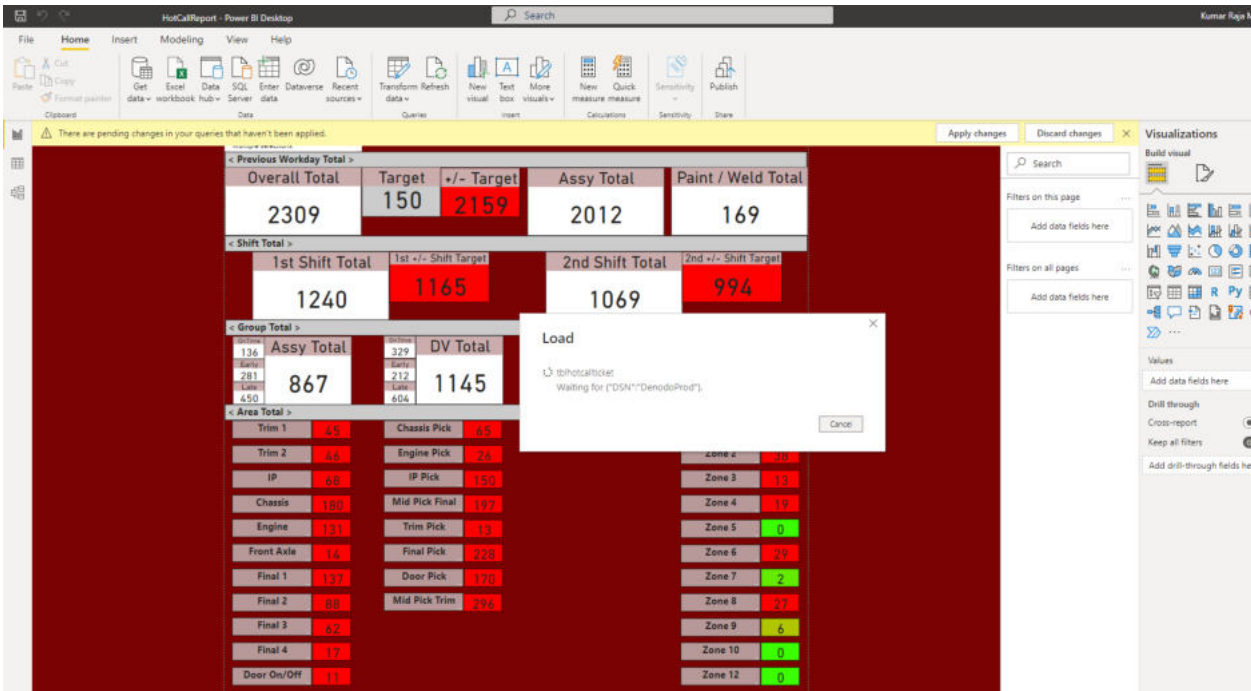
```
Source = Denodo.Contents("DenodoProd", null, []),
tmna_tbdp_Database = Source{[Name="tmna_tbdp", Kind="Database"]}[Data],
mfg_parts_hotcall_trn =
tmna_tbdp_Database{[Name="mfg_parts_hotcall_trn"]}[Data],
tblhotcallticket1 = mfg_parts_hotcall_trn{[Name="tblhotcallticket"]}[Data],
#"BusEntryCd Filter" = Table.SelectRows(tblhotcallticket1, each [busentrycd] = "04"),
Last3MonthFilter = Table.SelectRows(#"BusEntryCd Filter", each
Date.IsInPreviousNMonths([createddt], 3) or Date.IsInCurrentMonth([createddt]))
in
    Last3MonthFilter
```



f. Click on Done



g. click on Close & Apply



Note : Update All Redshift Tables connection string values in the dashboard by following steps mentioned above from D to G

Publish updated Power BI Report to Gateway:

In Power BI Desktop, choose File > Publish > Publish to Power BI (Refer below screenshot)

HotCallReport - Power BI Desktop

Search

FileHomeInsertModelingViewHelp

PasteCutCopyFormat painterClipboard

Get dataExcelData SQL ServerEnter dataRecent sourcesData

Transform Refresh dataQueries

New visualText boxMore visualsInsert

New Quick measure measureCalculations

SensitivitySensitivityShare

Publish

Hot Call Previous Date Detail

TOYOTA | MS

Last Refresh: 8/10/2022 10:22:21 AM

Filters

Search

Filters on this page

Add data fields here

Filters on all pages

Add data fields here

TicketID	ShiftDate	Shift	Supplier	Dock	PalletCode	KBN	PartNum	PartDesc	VehiclestoEmpty	CallBoxes	CallPieces	CallTime	SeqNum	BodyNo
HCM-TMMMS-29529	5/1/2022	2	91400	J1	S	0906	314201209000	CYL. MASTER L/RESVR	768	0	6	5/2/2022 10:41:43 AM	310	18458
HCM-TMMMS-29530	5/1/2022	2	14060	J1	M	J1T9	900801169900	BOLT. WASHER	1	0	25	5/2/2022 10:43:43 AM	309	18443
HCM-TMMMS-29531	5/1/2022	2	25356	J1	A	JPT0	7348002530C0	BELT RR IN W/CTR. LH	20	2		5/2/2022 10:43:48 AM	234	18379
HCM-TMMMS-29532	5/1/2022	2	05470	J1	J	JNL4	6211202400C0	BOARD CWL SD TRIM L	17	0	17	5/2/2022 10:45:47 AM	234	18379
HCM-TMMMS-29533	5/1/2022	2	14060	J1	S2	J564	90182A003400	NUT. W/WASHER	47	0		5/2/2022 10:55:51 AM	265	18408
HCM-TMMMS-29534	5/1/2022	2	21710	J1	Q	J502	90119A035100	BOLT. W/WASHER	0	0		5/2/2022 10:56:29 AM	265	18408
HCM-TMMMS-29535	5/1/2022	2	91400	J1	S	0212	909500195400	PLUG. HOL					265	18408
HCM-TMMMS-29536	5/1/2022	2	21710	J1	C	J515	90119A044000	BOLT. W/M					265	18408
HCM-TMMMS-29537	5/1/2022	2	21710	J1	S3	J4J3	90105A010500	BOLT. FLAP					265	18408
HCM-TMMMS-29538	5/2/2022	1	14060	J1	D	J1T9	900801169900	BOLT. WAS					220	18364
HCM-TMMMS-29539	5/2/2022	1	14060	J1	D	J7K0	90119A024700	BOLT. W/M					215	18358
HCM-TMMMS-29540	5/2/2022	1	01508	J1	JX	JYU5	82115F2X4000	WIRE. ENG					310	18458
HCM-TMMMS-29541	5/2/2022	1	01508	J1	JX	JYU6	8211602Y4000	WIRE. ENG					310	18458
HCM-TMMMS-29542	5/2/2022	1	14060	J1	D	J7K0	90119A024700	BOLT. W/M					213	18357
HCM-TMMMS-29543	5/2/2022	1	01508	J1	JZ	JYU6	82115F2X5000	WIRE. ENG					310	18458
HCM-TMMMS-29544	5/2/2022	1	21710	J1	D	J867	90151A000300	SCREW.XR					211	18356
HCM-TMMMS-29545	5/2/2022	1	31520	J1	D	JHL4	90159A001800	SCREW.W/					202	18345
HCM-TMMMS-29546	5/2/2022	1	21710	J1	D	J156	900801507400	SCREW. WASHER	40	0		5/2/2022 11:29:04 AM	204	18348
HCM-TMMMS-29547	5/2/2022	1	19514	J1	V	JLD6	164A20T03000	TANK. RSRV SEAL LCAP	22	0	22	5/2/2022 11:29:04 AM	329	18481
HCM-TMMMS-29548	5/2/2022	1	19086	J1	J	JNF2	611650202000	BRKT IP CWL SD R	22	0	22	5/2/2022 11:29:51 AM	329	18481
HCM-TMMMS-29549	5/2/2022	1	58135	J1	B	JMN7	550860257000	DUCT SA HT REG CTR	12	1		5/2/2022 11:35:09 AM	315	18463
HCM-TMMMS-29550	5/2/2022	1	19505	J1	A	JLH3	177000T35000	CLNR. ASY AIR W ELMT	7	2		5/2/2022 11:40:19 AM	226	18368
HCM-TMMMS-29551	5/2/2022	1	31561	J1	T1	J556	8791002K90J0	MIR. ASY OUT RR VW R	0	0	0	5/2/2022 11:41:54 AM	245	18390
HCM-TMMMS-29552	5/2/2022	1	21710	J1	Q2	J528	90119A045700	BOLT. W/WASHER	0	0		5/2/2022 11:44:30 AM	288	18431
HCM-TMMMS-29553	5/2/2022	1	17651	J1	RK	JX63	7585002908B2	MLDG. BDY PNL SET	0	0	0	5/2/2022 11:46:28 AM	220	18364

Publishing to Power BI

Success!

Open 'HotCallReport.pbix' in Power BI

Get Quick Insights

Did you know?

You can create a portrait view of your report, tailored for mobile phones. On the View tab, select Mobile Layout. [Learn more](#)

Got it

Hot Call Report-Previous Day

Hot Call Report-History

Previous Date Detail

History Detail

←

→

🔄

🏠

https://app.powerbi.com/groups/52031b7f-b5b7-4ae4-ba67-a5d0067a2692/reports/24c31d02-3334-4614-b296-7ebaa14e8c59/ReportSectionbc148c0b596d3c21bef8

Would you like to set Microsoft Edge as your default browser?

Set as default

Power BI

Denodo Testing

HotCallReport | Data updated 8/18/22

3 days left

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Workspaces

Denodo Testing

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Hot Call Report-Previou...

Hot Call Report-History

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Subscribe

Edit

We updated the look of reports

Take a tour, and we'll show you how to get around.

Hot Call Previous Date Detail

Last Refresh:

TicketID	ShiftDate	Shift	Supplier	Dock	PalletCode	KBN	PartNum	PartDesc	VehiclestoEmpty	CallBoxes	CallPieces	CallTime
HCM-TMMMS-29529	5/1/2022	2	91400	J1	S	0906	314201209000	CYL MASTER L/RESVR	768	0	6	5/2/2022 10:41:43
HCM-TMMMS-29530	5/1/2022	2	14060	J1	M	J1T9	900801169900	BOLT, WASHER	1	0	25	5/2/2022 10:43:43
HCM-TMMMS-29531	5/1/2022	2	25356	J1	A	JPT0	7348002530C0	BELT RR IN W/CTR, LH	20	2		5/2/2022 10:43:48
HCM-TMMMS-29532	5/1/2022	2	05470	J1	J	JNL4	6211202400C0	BOARD CWL SD TRIM L	17	0	17	5/2/2022 10:45:47
HCM-TMMMS-29533	5/1/2022	2	14060	J1	S2	JS64	90182A003400	NUT, W/WASHER	47	0		5/2/2022 10:55:51
HCM-TMMMS-29534	5/1/2022	2	21710	J1	Q	JS02	90119A035100	BOLT, W/WASHER	8	0		5/2/2022 10:56:29
HCM-TMMMS-29535	5/1/2022	2	91400	J1	S	0212	909500195400	PLUG, HOLE	31	0		5/2/2022 10:57:06
HCM-TMMMS-29536	5/1/2022	2	21710	J1	C	JS15	90119A044000	BOLT, W/WASHER	17	0		5/2/2022 10:57:33
HCM-TMMMS-29537	5/1/2022	2	21710	J1	S3	J4J3	90105A010500	BOLT, FLANGE	114	0		5/2/2022 10:58:08
HCM-TMMMS-29538	5/2/2022	1	14060	J1	D	J1T9	900801169900	BOLT, WASHER	196	1		5/2/2022 11:11:05
HCM-TMMMS-29539	5/2/2022	1	14060	J1	D	J7K0	90119A024700	BOLT, W/WASHER	200	1		5/2/2022 11:13:48
HCM-TMMMS-29540	5/2/2022	1	01508	J1	JX	JYU5	82115F2X4000	WIRE, ENG ROOM MAIN	393	0	2	5/2/2022 11:14:05
HCM-TMMMS-29541	5/2/2022	1	01508	J1	JX	JYLV	821160ZY4000	WIRE, ENG ROOM MAIN	351	0	2	5/2/2022 11:14:42
HCM-TMMMS-29542	5/2/2022	1	14060	J1	D	J7K0	90119A024700	BOLT, W/WASHER	200	1		5/2/2022 11:15:16
HCM-TMMMS-29543	5/2/2022	1	01508	J1	JZ	JYU6	82115F2X5000	WIRE, ENG ROOM MAIN	8	0	4	5/2/2022 11:15:25
HCM-TMMMS-29544	5/2/2022	1	21710	J1	D	J867	90151A000300	SCREW,XRECS COUNTER	160	0		5/2/2022 11:17:41
HCM-TMMMS-29545	5/2/2022	1	31520	J1	D	JHL4	90159A001800	SCREW,W/WASH TAPPING	97	0		5/2/2022 11:18:28
HCM-TMMMS-29546	5/2/2022	1	21710	J1	D	J1S6	900801507400	SCREW, WASHER TAP	46	0		5/2/2022 11:22:52
HCM-TMMMS-29547	5/2/2022	1	19514	J1	V	JLD6	164A20T03000	TANK, RSRV SEAL LCAP	22	0	22	5/2/2022 11:29:04
HCM-TMMMS-29548	5/2/2022	1	19086	J1	J	JNF2	611650202000	BRKT IP CWL SD R	22	0	22	5/2/2022 11:29:51
HCM-TMMMS-29549	5/2/2022	1	58135	J1	B	JMN7	550860257000	DUCT SA HT REG CTR	12	1		5/2/2022 11:35:09
HCM-TMMMS-29550	5/2/2022	1	19505	J1	A	JLH3	177000T35000	CLNR ASY AIR W ELMT	7	2		5/2/2022 11:40:19
HCM-TMMMS-29551	5/2/2022	1	31561	J1	T1	JS56	8791002K90J0	MIR ASY OUT RR VW R	0	0	0	5/2/2022 11:41:54
HCM-TMMMS-29552	5/2/2022	1	21710	J1	Q2	JS28	90119A045700	BOLT, W/WASHER	0	0		5/2/2022 11:44:30
HCM-TMMMS-29553	5/2/2022	1	17651	J1	RK	JX63	758500290882	MLDG BDY PNL SET	0	0	0	5/2/2022 11:46:28

3. Connecting Tableau to Denodo

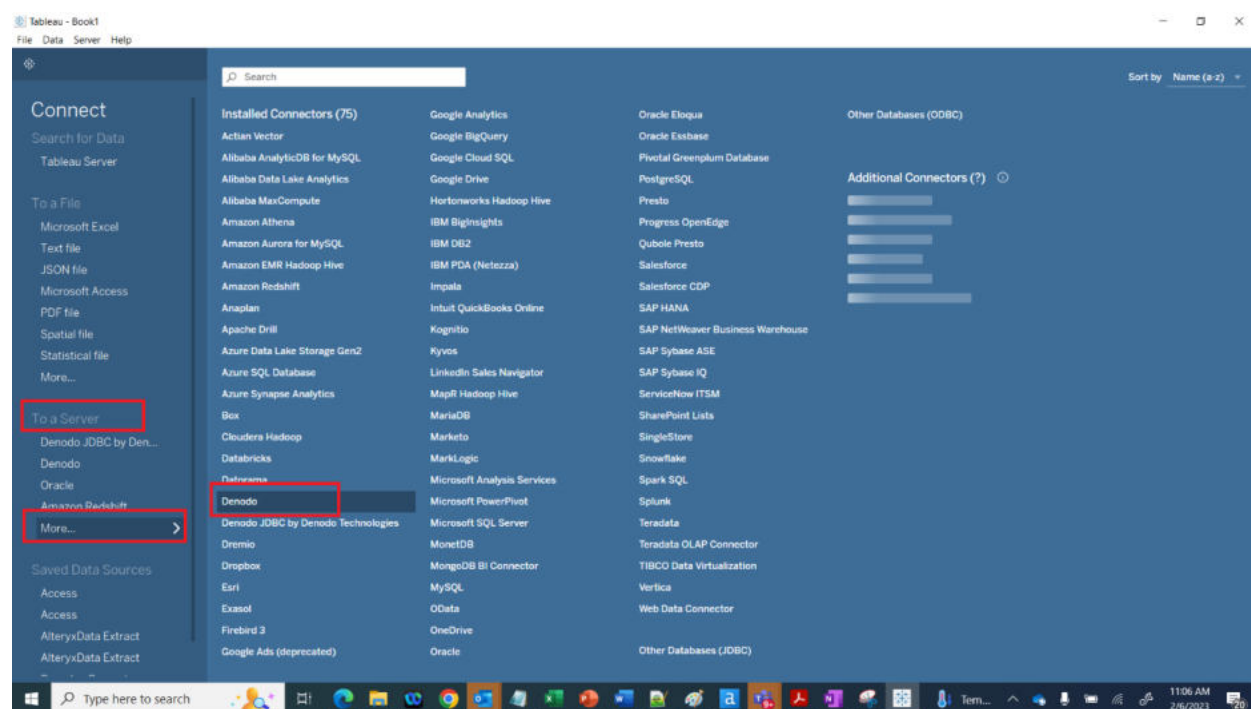
1. Tableau and Denodo integration

- a. Close all Tableau Desktop instances
- b. Copy the file 'denodo_native_odbc.tdc' to ...\\Documents\\My Tableau Repository\\Datasources and update the client secret key value as per the attached link below.

[denodo_native_odbc.tdc](#)

[Process to Update Denodo Client Secret in the Tableau Custom connection file.docx](#)

- c. Start Tableau Desktop
- d. Under Connect option, Navigate to 'To a server >> More' and select 'Denodo' /serach for 'Denodo'



- e. Enter below details in the connection dialog box and click on 'Sign In' button

Server: data.prod.tbdp.toyota.com

Port: 9996

Database: tmna_tbdp

Authentication: Username & Password

Username: Enter your email

Password: Enter Active Directory/Network password

Require SSL: Checked

Denodo

General

Initial SQL

Server

data.prod.tbdp.toyota.com

Port

9996

Database

tmna_tbdp

Authentication

Username and Password

Username

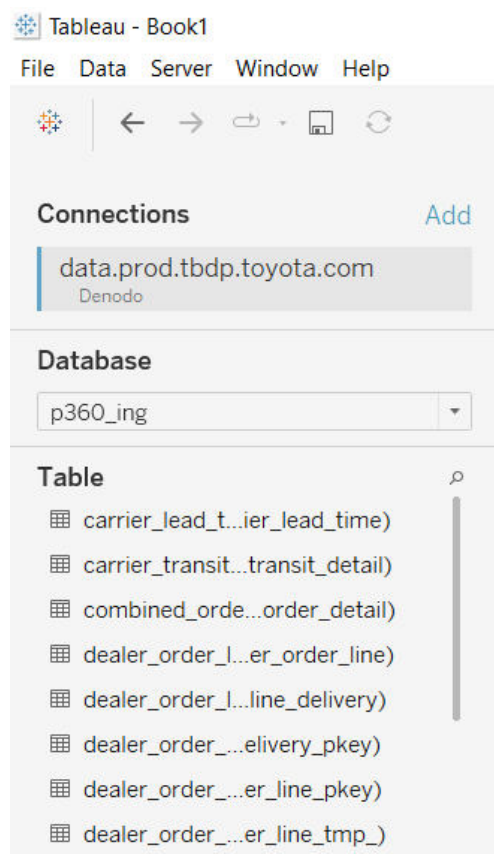
Password

Optional

☐ Require SSL

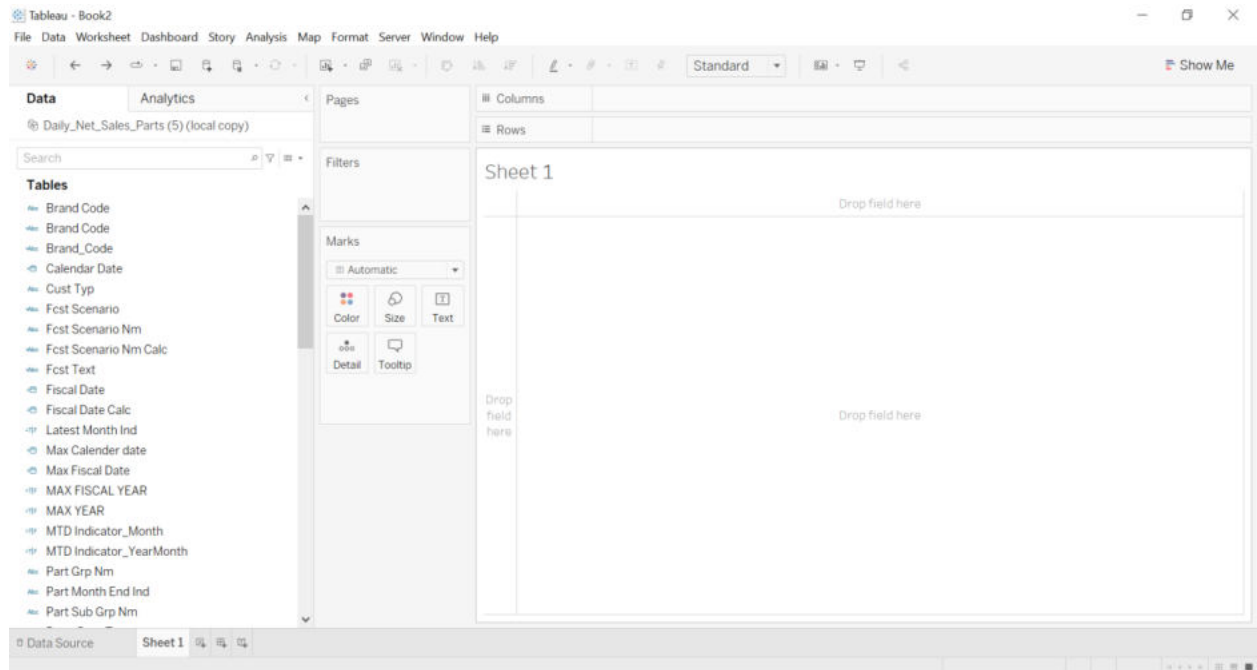
Sign In

- f. Once connected, select the Schema(Database) you would like to connect to, Select the table from the schema and derive insights

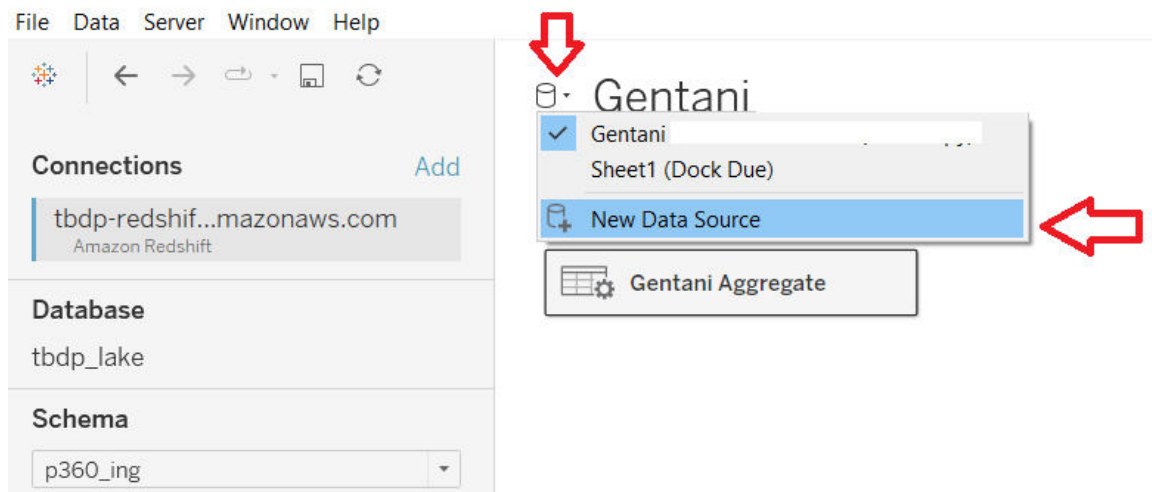


Steps to replace Redshift Data Source with Denodo Data Source

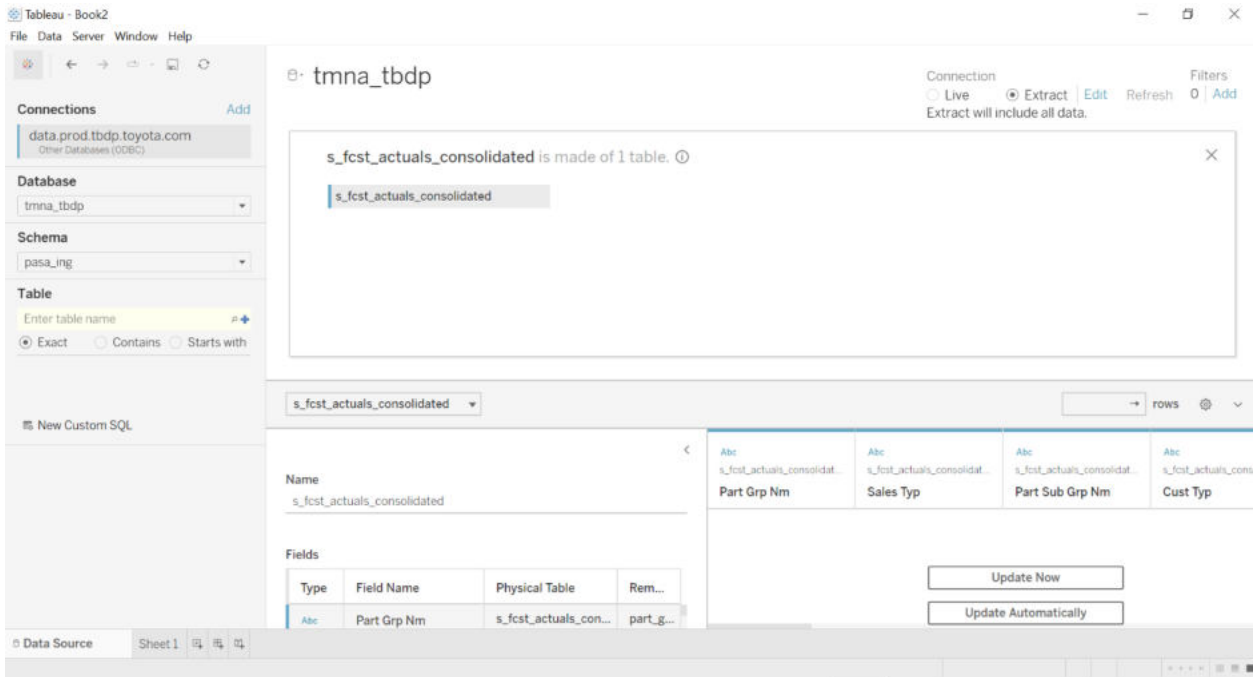
1. Open the Redshift connected data source in Tableau Desktop



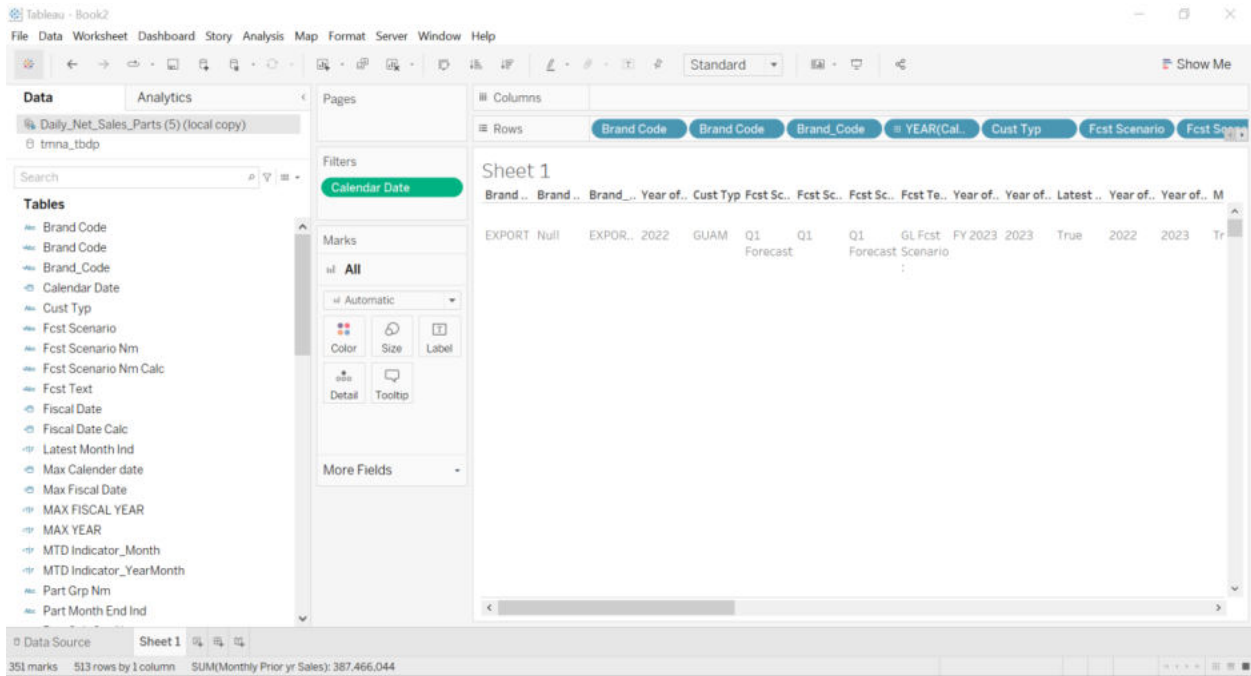
2. Navigate to the 'Data Source' page (bottom left).
3. Add New Data source and Connect to Denodo data source using the steps given above



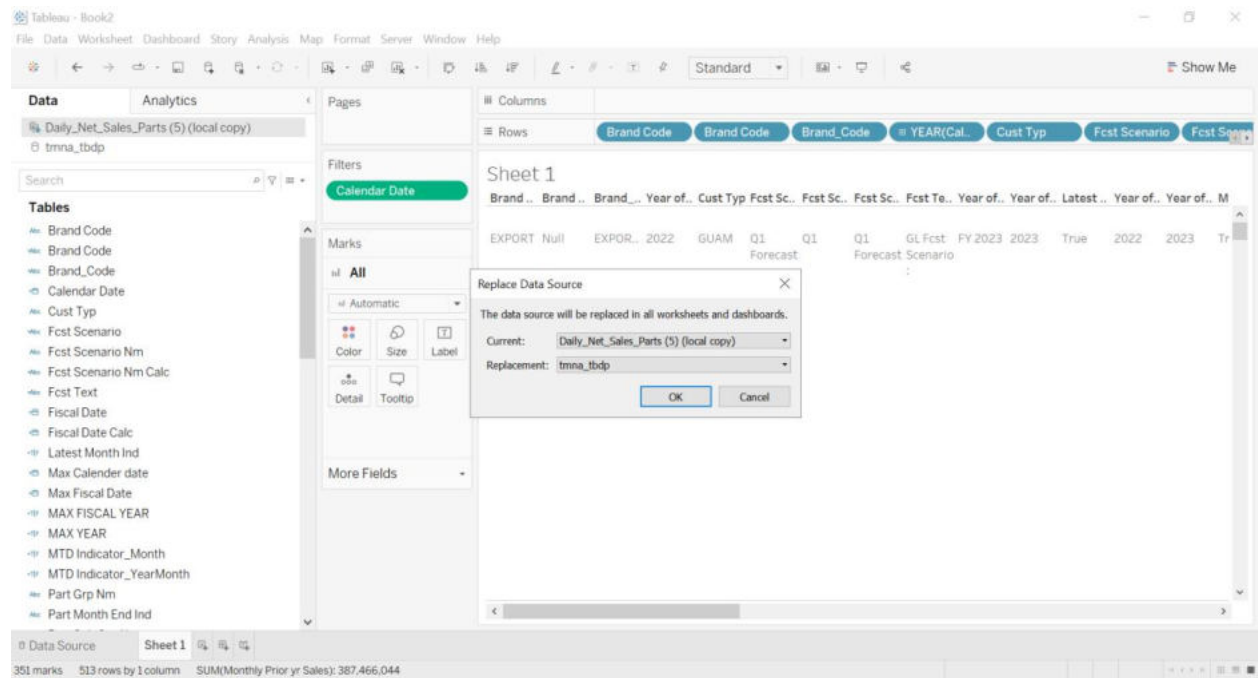
4. In Data source page, drag and drop the required table after selecting schema. Change the connection to Live/Extract accordingly.



5. Now,come to Sheet 1, put a simple filter and drag & drop all fields from the primary data source to the view.



6. Now,replace data source by clicking on *Data > Replace Data Source*. Select the current Data source (Redshift) and replacement data source (Denodo) and click on OK button.



- 7. After replacing data source, check if all the fields are working fine, clear the page and publish to server

4. Connecting Alteryx to Denodo

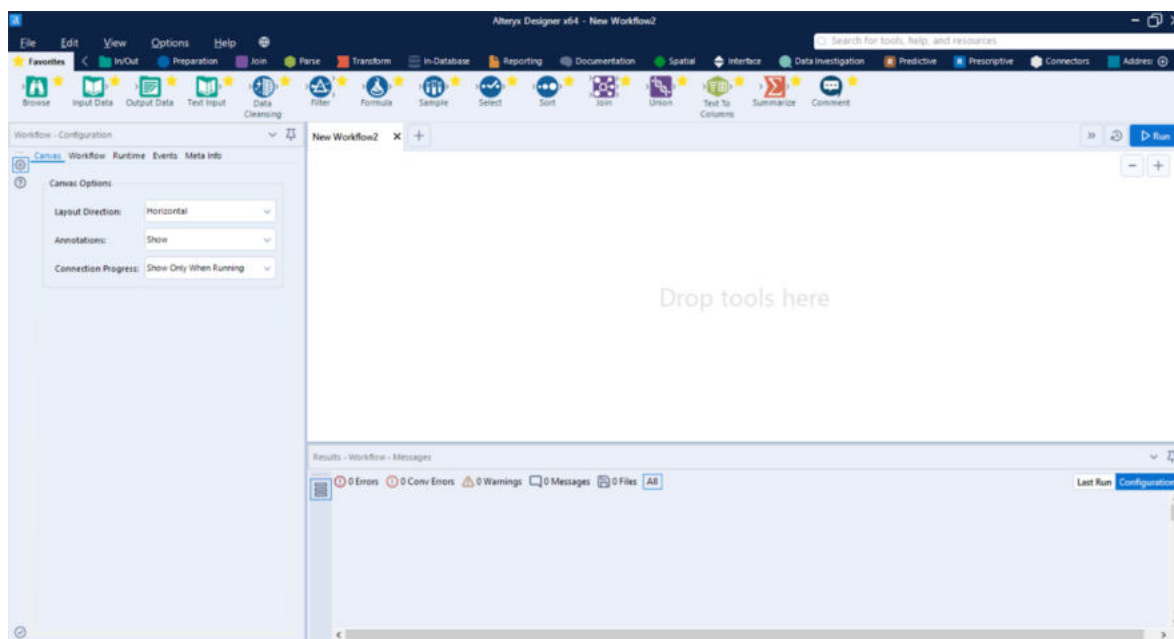
There are different ways you can connect to Denodo from Alteryx (same as connection to Redshift). If you want to publish the workflow to Alteryx Gallery, you must configure your workflow to use either Gallery based connection or INDBC file based connection.

In order to create a Gallery based Data connection (Standard tools), the Alteryx Admin team will require a Service account. Please submit a Data Access Request (DAR) request to get the Service account created.

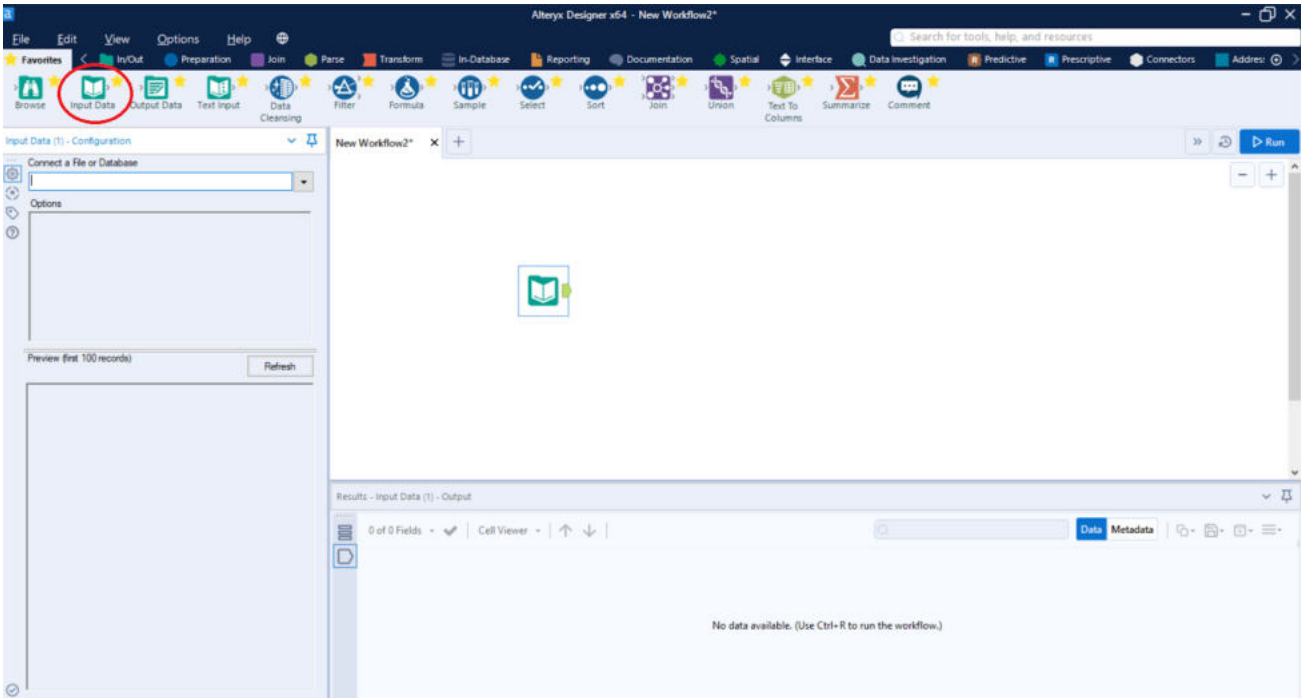
Connection Option#1 : Connection using Standard tools

1.1 Connection using Local ODBC DSN (will work locally only)

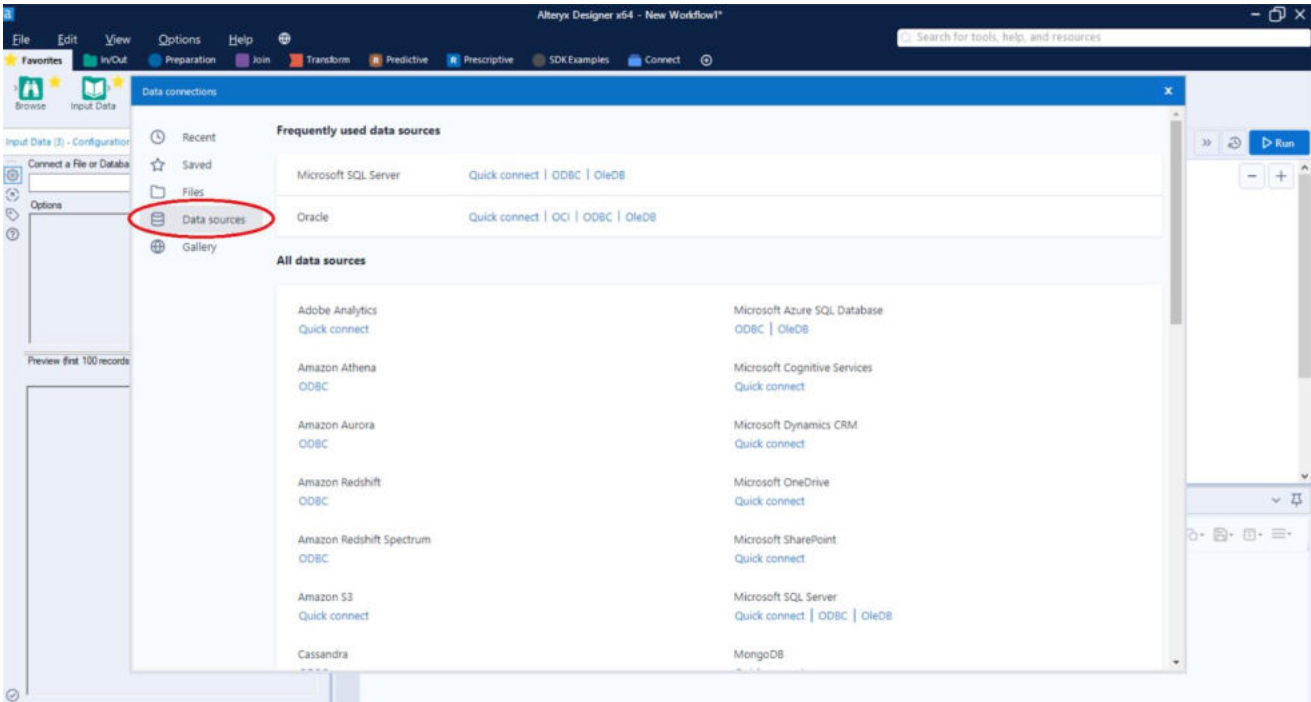
- a. Open the Alteryx Designer Tool



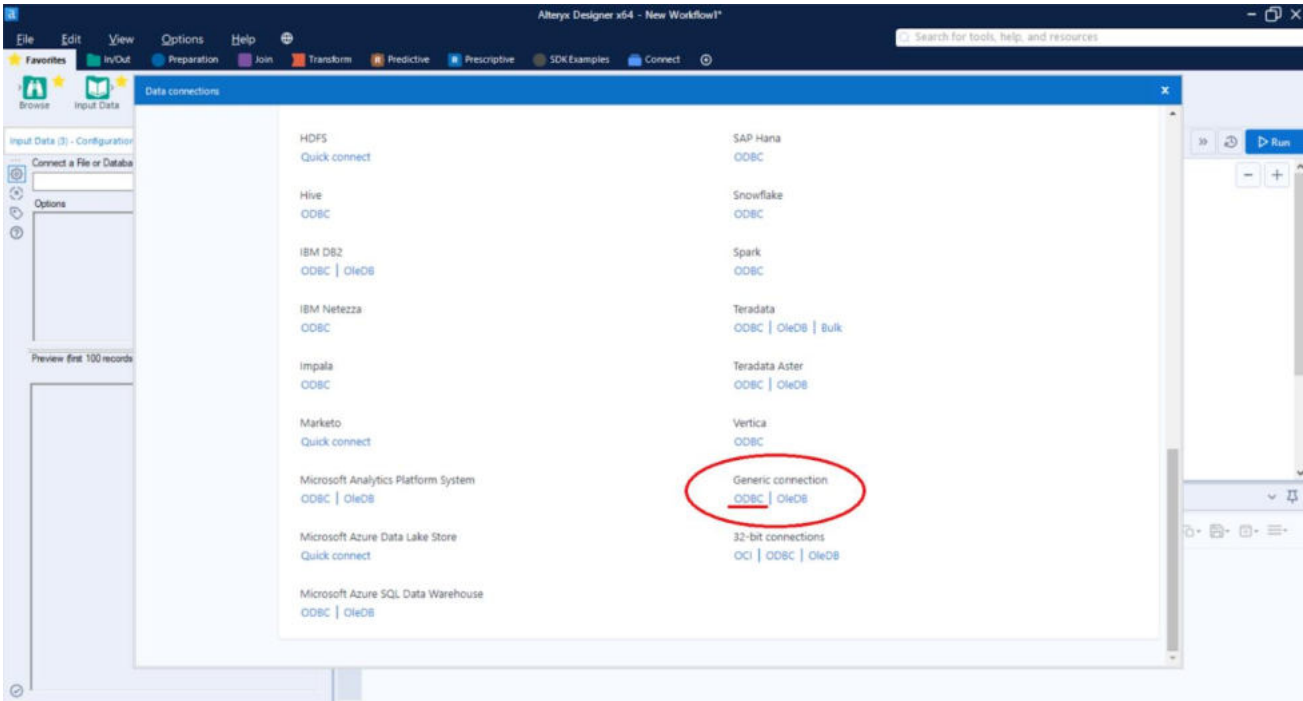
- b. Drag and drop the Input Data tool from the toolbox into the canvas.



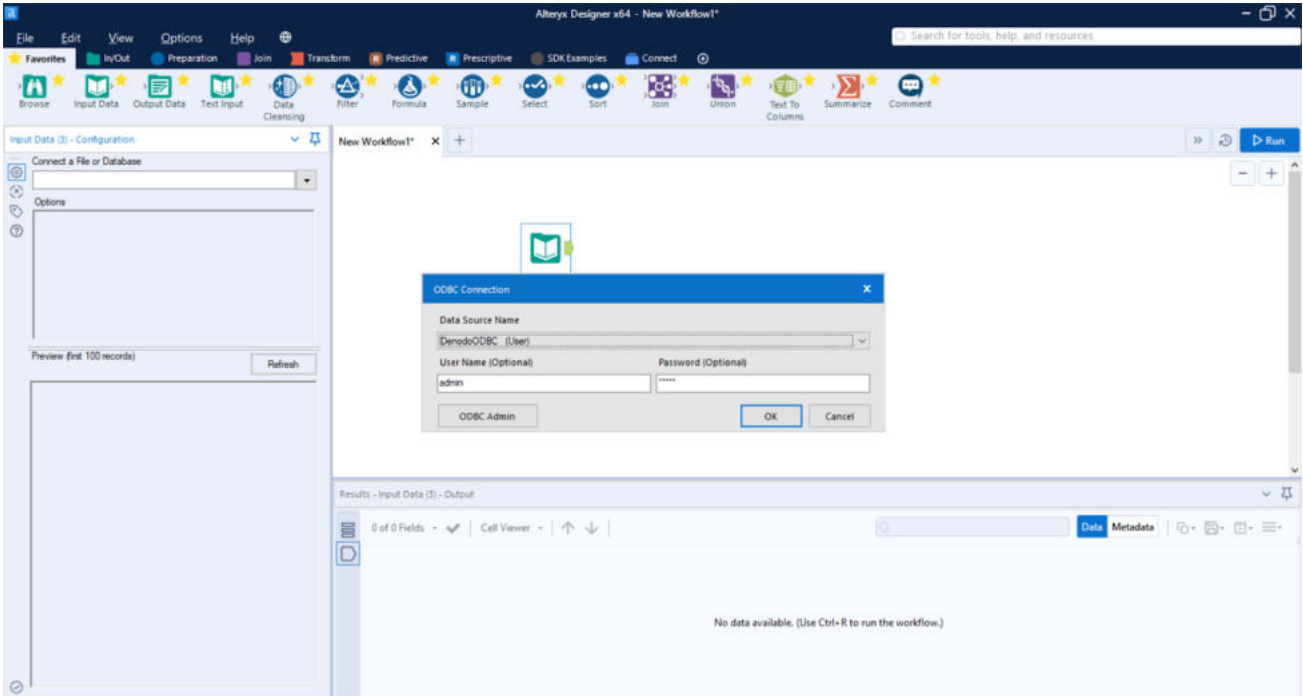
- c. Click on drop down menu for Connect a File or Database and click on Data Sources on the pop-up window.



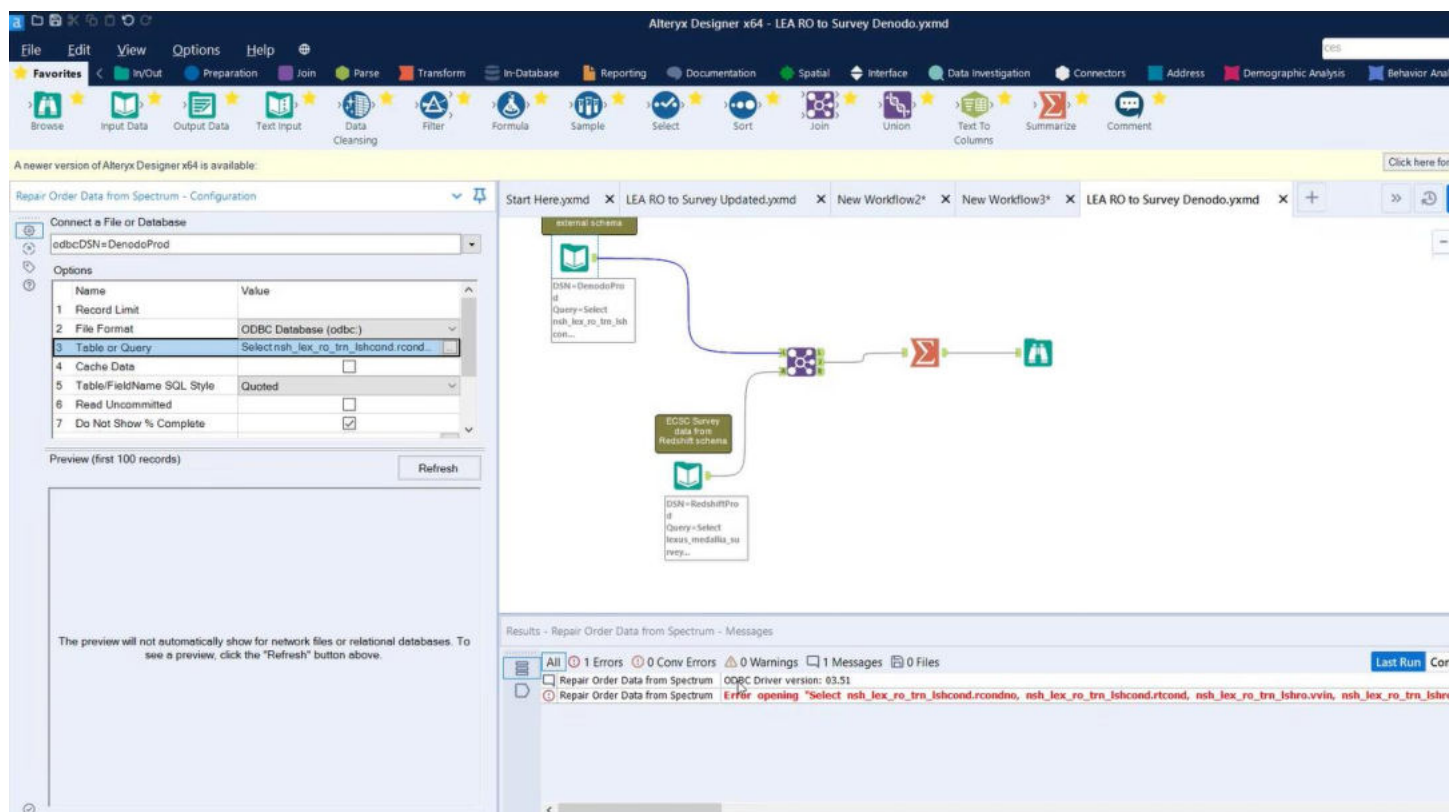
- d. Select ODBC on Generic Connection



- e. In the Data Source Name drop down menu select the DSN you set up in the initial step, type User Name and Password then click OK.

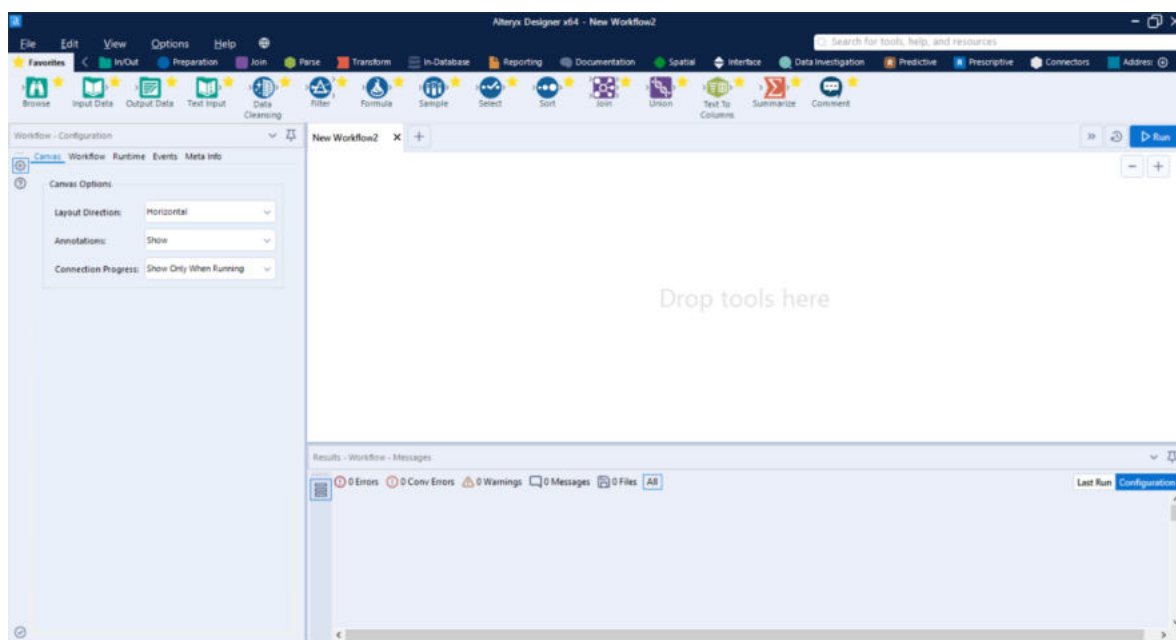


- f. Drag and drop the Denodo views to the left side, you can also use the option SQL Editor. To do that, select the required fields, edit them and then click Ok.

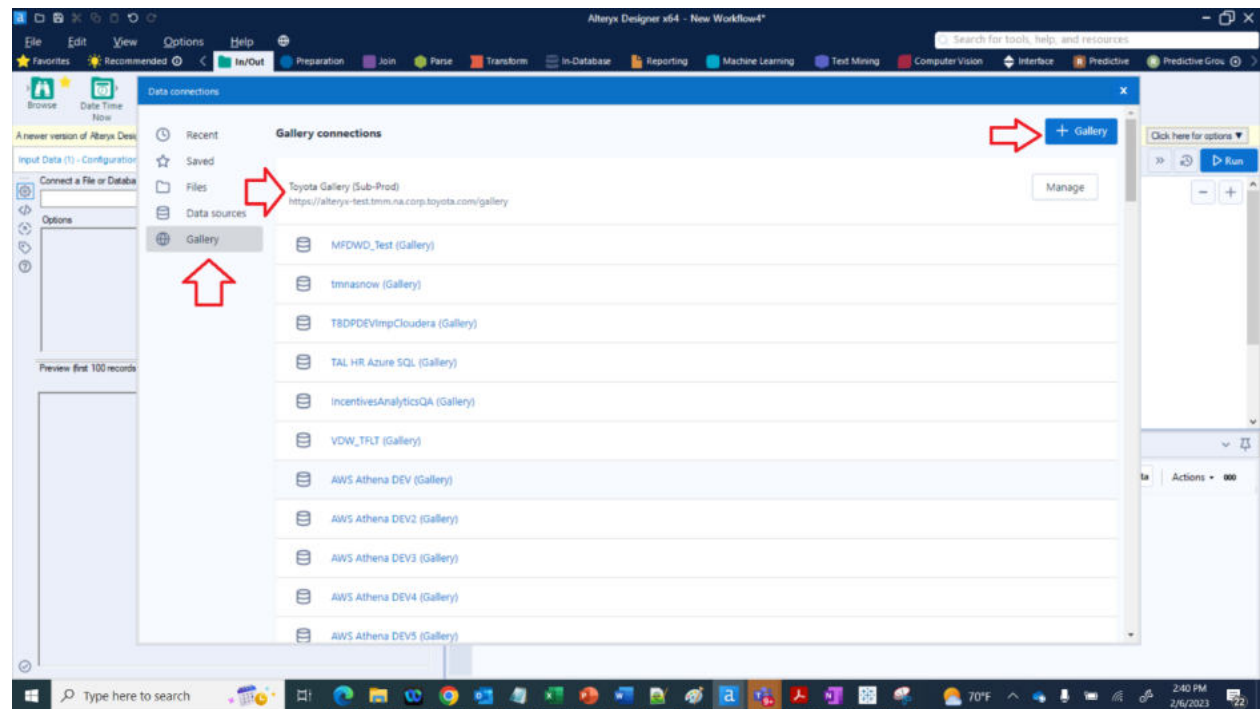


Connection using Alteryx Gallery Connection (Can be used only with Standard tools)

1. Open the Alteryx Designer Tool



2. Drag and drop the Input Data tool from the toolbox into the canvas
3. Click on drop down menu for **Connect a File or Database** option and navigate to 'Gallery' section.



4. Select the relevant Denodo connection from the list. If you don't see the desired connection listed, please reach out to Analytics COE AnalyticsCOE@internal.toyota.com
5. If you don't see the TMNA Alteryx Gallery listed under the Gallery Connections section, you can add the Gallery using '+Gallery' button on the top-left corner

Gallery URL:

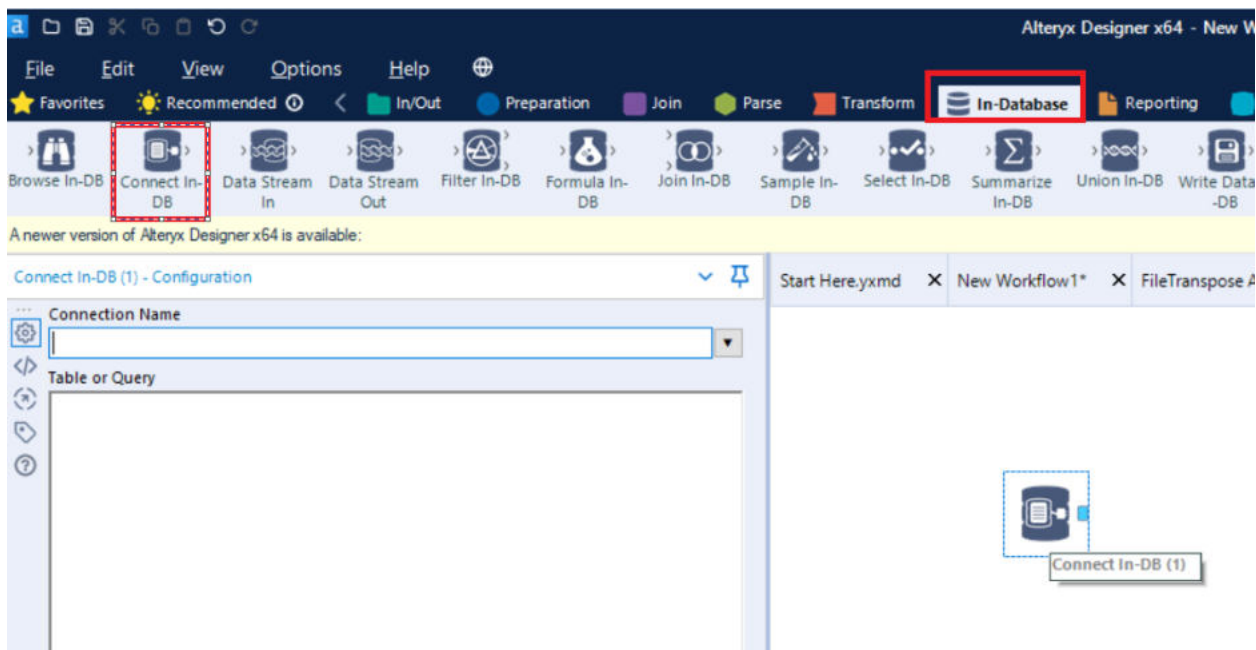
TEST: <https://alteryx-test.tmm.na.corp.toyota.com/gallery>

PROD: <https://alteryx.tmm.na.corp.toyota.com/gallery>

6. The Choose Table or Specify Query window opens.
7. Switch to **Visual Query Builder** tab. Select the **Tables** and drag them on the canvas as needed or you can write a SQL using **SQL Editor** window

Connection Option#2 : Connection using In-DB tools

1. Open **Alteryx Designer** and Add a **Connect In-DB** tool on the canvas

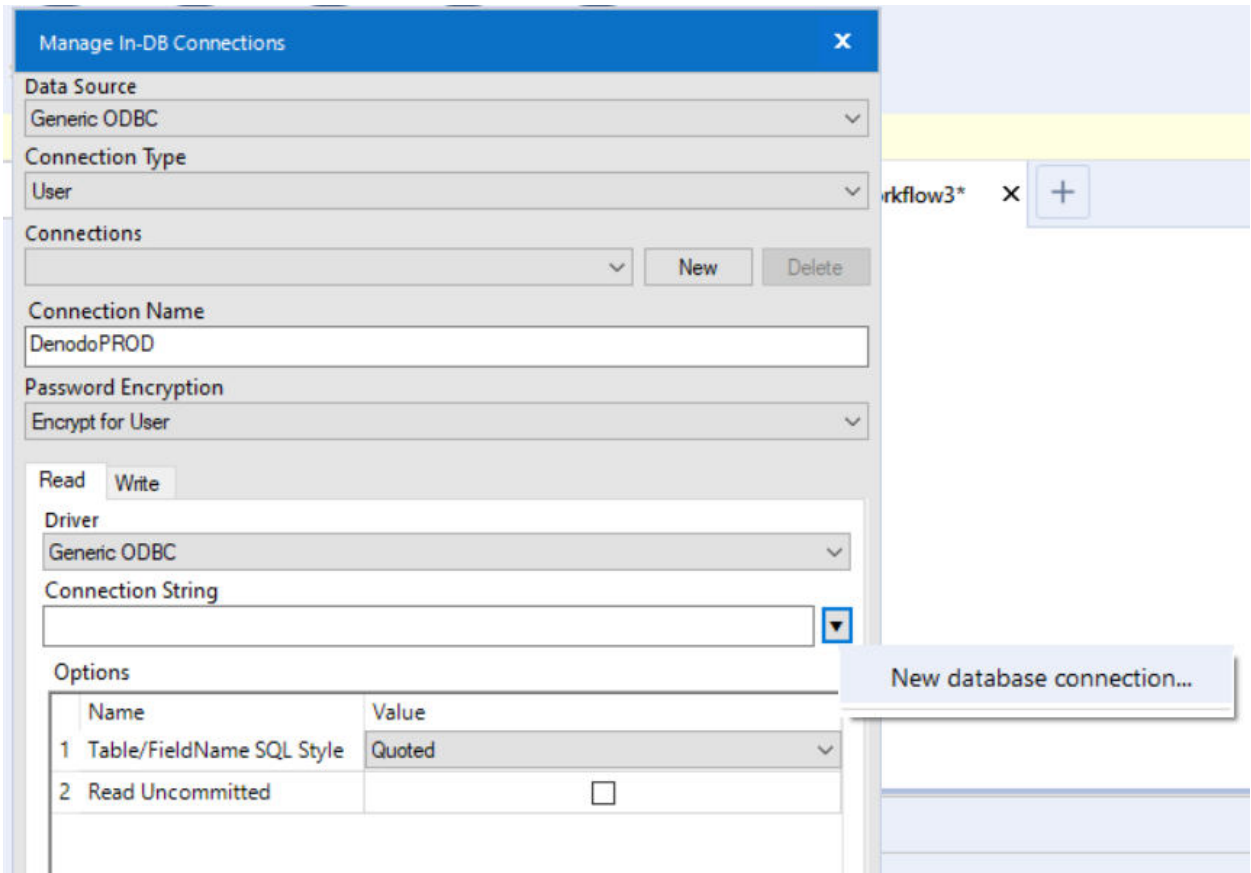


2. If you already have the shared .indbc file created, paste the file path in the Connection Name windows in the format '*File:<Path to .indbc file>*' , click on the Canvas. The **Choose Table or Specify Query** window will open in few seconds. Jump to Step#18.
If you need to create new connection, follow the steps below
3. To create new connection, Select the **Connection Name** dropdown arrow and select **Manage Connections**.
4. Select **Data Source** and select **Generic ODBC** as Data source.
5. Select **Connection Type** and select type.
 - **User:** Create a connection that only you can use.
 - **File:** Saves a database connection as an .indbc file so it can be packaged with a workflow. If this option is selected, a **Connection File** path location must be specified in order to save the file.

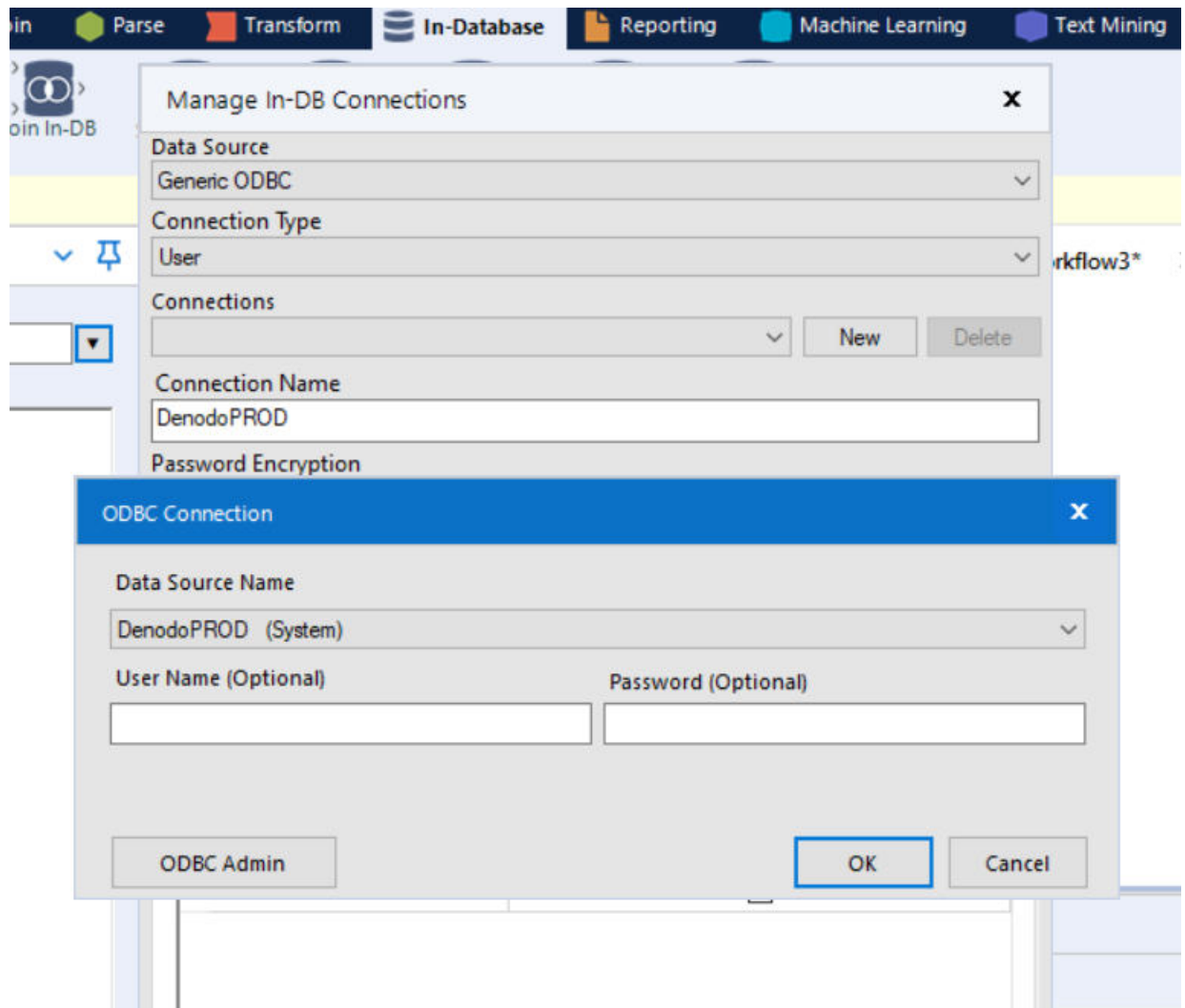
Note: If you are going to publish the workflow to the Gallery, use the File option and place the indbc file on the Alteryx Shared drive.

If you need access to/need help updating an existing Service account based indbc file created for your application, please reach out to Analytics COE team Analytics COE <AnalyticsCOE@internal.toyota.com>

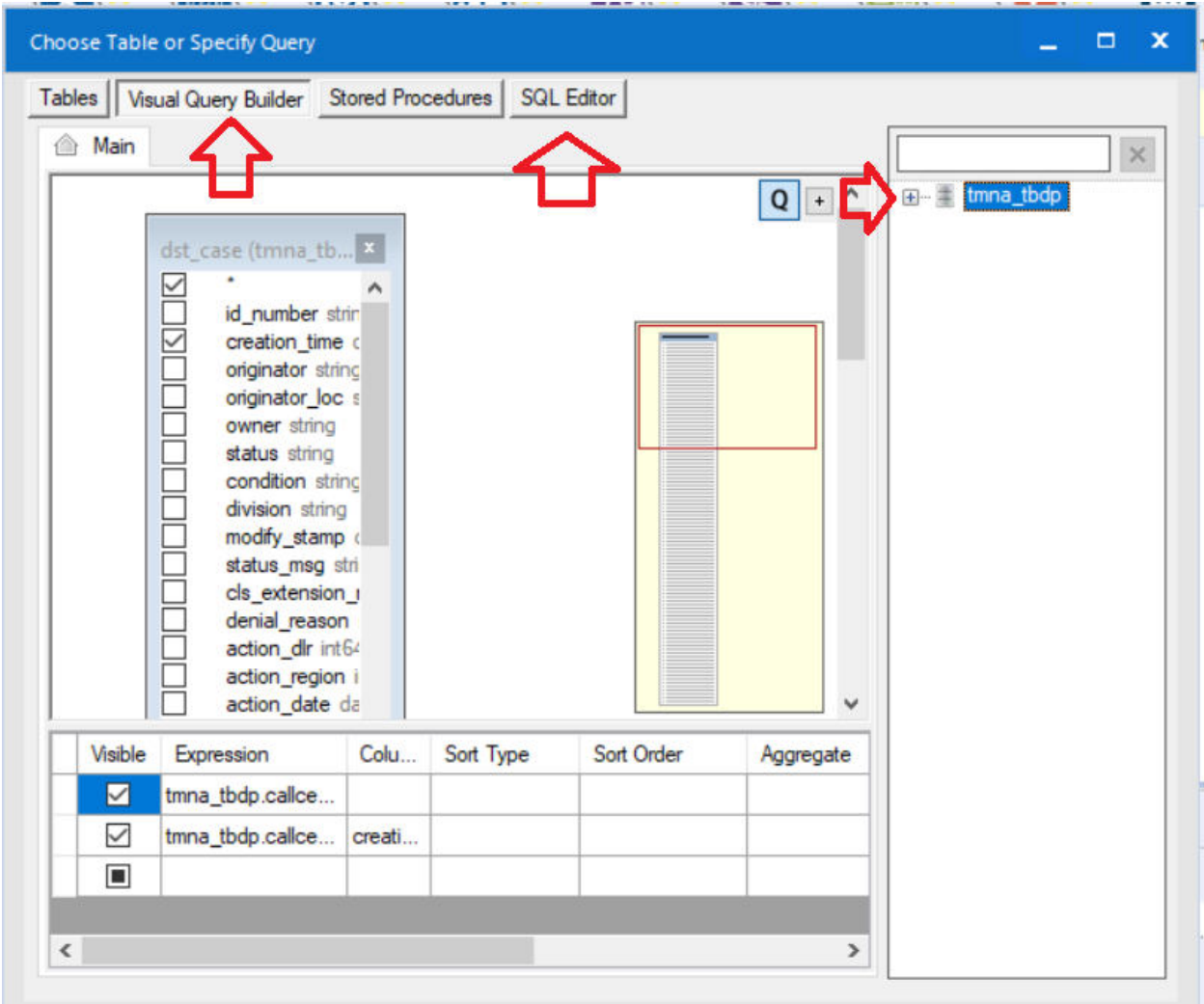
- 6. Select **Connections** and select **New**.
- 7. In **Connection Name**, enter a name for the connection.
- 8. Select **Password Encryption** and select an encryption option:
 - **Hide**: Hide the password using minimal encryption.
- 9. On the **Read** tab, leave **Driver** option as default
- 10. Select the **Connection String** dropdown arrow and select **New database connection**.



- 11. Select the Denodo ODBC DSN, enter your credentials and click on 'OK'



12. Select the **Write** tab.
13. Select **Driver** and make sure its set to '*Same as Read Driver*'
14. **Connection String** will be automatically populated
15. Click on **Apply** button
16. Select **OK**.
17. The **Choose Table or Specify Query** window will open in few seconds.
18. Switch to **Visual Query Builder** tab. Select the **Tables** and drag them on the canvas as needed or you can write a SQL using **SQL Editor** window



19. Test the SQL

20. Click **OK**

Note : Please check and update any Redshift custom specified queries to Denodo compatible format (Refer below sample query)

Redshift Query:

Select nsh_lex_ro_trn_lshcond.rcondno, nsh_lex_ro_trn_lshcond.rtcond, nsh_lex_ro_trn_lshro.vvin, nsh_lex_ro_trn_lshro.rdte, nsh_lex_ro_trn_lshro.rdlrno, nsh_lex_ro_trn_lshro.rorderno
From svc_history_lexus_us_trn_ext.lshro nsh_lex_ro_trn_lshro Inner Join
svc_history_lexus_us_trn_ext.lshcond nsh_lex_ro_trn_lshcond On nsh_lex_ro_trn_lshro.rdlrno =
nsh_lex_ro_trn_lshcond.rdlrno And nsh_lex_ro_trn_lshro.rorderno = nsh_lex_ro_trn_lshcond.rorderno


```
And nsh_lex_ro_trn_lshro.rdte = nsh_lex_ro_trn_lshcond.rdte
where substring (nsh_lex_ro_trn_lshro.rdte FROM (len(nsh_lex_ro_trn_lshro.rdte) - 3) FOR 4) = '2019'
```

Denodo Query:

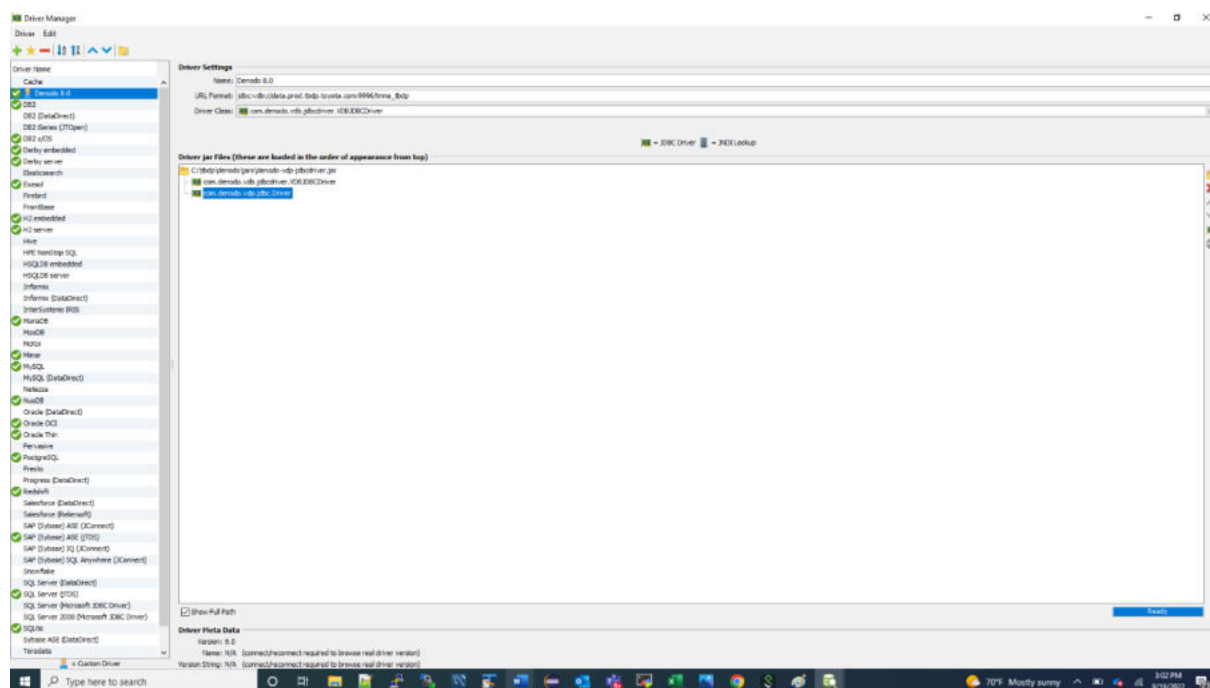
```
Select nsh_lex_ro_trn_lshcond.rcondno, nsh_lex_ro_trn_lshcond.rtcond, nsh_lex_ro_trn_lshro.vvin,
nsh_lex_ro_trn_lshro.rdte, nsh_lex_ro_trn_lshro.rdlrno, nsh_lex_ro_trn_lshro.rorderno
From svc_history_lexus_us_trn_ext.lshro nsh_lex_ro_trn_lshro Inner Join
svc_history_lexus_us_trn_ext.lshcond nsh_lex_ro_trn_lshcond On nsh_lex_ro_trn_lshro.rdlrno =
nsh_lex_ro_trn_lshcond.rdlrno And nsh_lex_ro_trn_lshro.rorderno = nsh_lex_ro_trn_lshcond.rorderno
And nsh_lex_ro_trn_lshro.rdte = nsh_lex_ro_trn_lshcond.rdte
where right( nsh_lex_ro_trn_lshro.rdte,4) = '2019'
```

Reference Link :

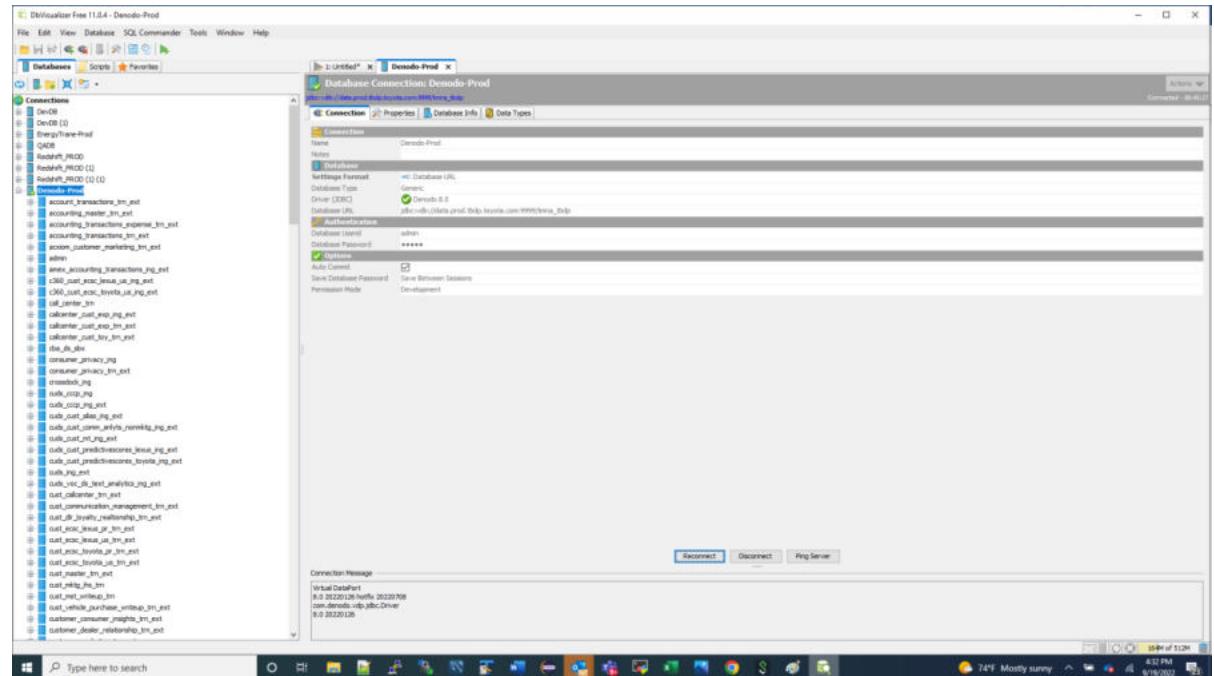
<https://community.denodo.com/answers/question/details?questionId=9067000000Xd8PAAS&title=Left+and+right+function+in+MSSQL>

5. Accessing Denodo using a SQL client(DbVisualizer)

1. Add the Denodo's JDBC driver ([Denodo JDBC JAR](#)) to the client application. To use the JDBC driver in your client, you have to add the .jar file to the classpath of your application.
2. In DBVisualizer, you have to go to **Tools > Driver Manager...** and in the **Driver Manager** window go to **Driver > Create Driver** and then browse to the VDP's driver file. Use the following driver settings and close the window to save the configuration:
 - a. **Name:** Denodo 8.0
 - b. **URL Format:** jdbc:vdb://data.prod.tbdp.toyota.com:9999/tmna_tbdp
 - c. **Driver Class:** com.denodo.vdp.jdbc.Driver



3. You have added the driver, you can configure a connection to your Denodo virtual database.
4. Go to **Database > Create Database Connection** and use the following settings for the connection:
 - **Driver (JDBC):** Denodo 8.0
 - **Database URL:** jdbc:vdb://data.prod.tbdp.toyota.com:9999/tmna_tbdp
 - **Database Userid:** Provide Toyota email
 - **Database Password:** Provide Toyota network password



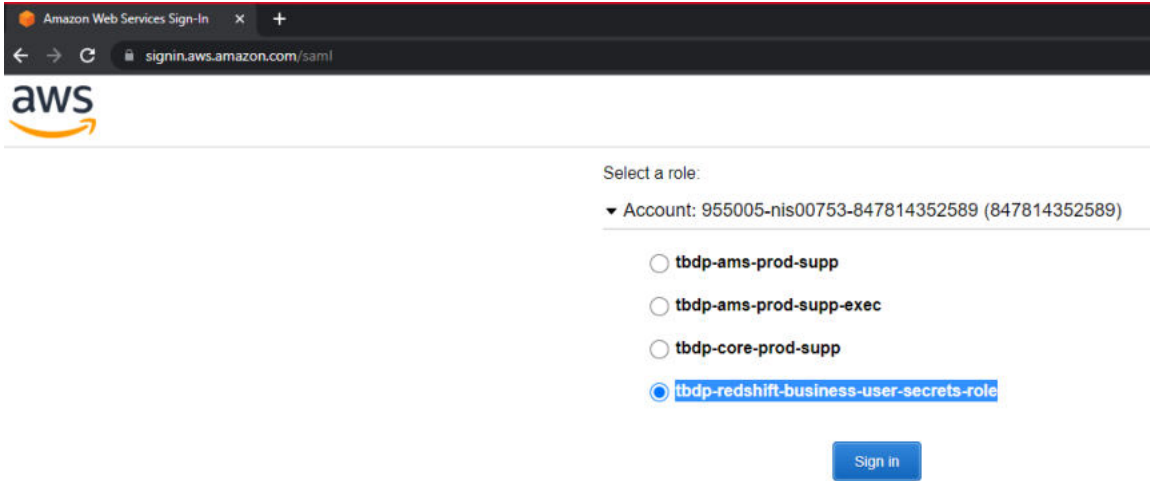
5. Including the OAuth Parameters in Driver Properties (please refer below screenshots)

- useOAuth2: true
- TokenEndPoint: <https://login.microsoftonline.com/014106b1-ea3a-48aa-88f8-58cbb5dde11/oauth2/v2.0/token>
- sslTrustServerCertificate: true
- scope: openid profile
- useIdToken: true
- oauthSSLVerify: true
- ssl: true
- **Please retrieve the clientID and clientSecret using below AWS URL.**

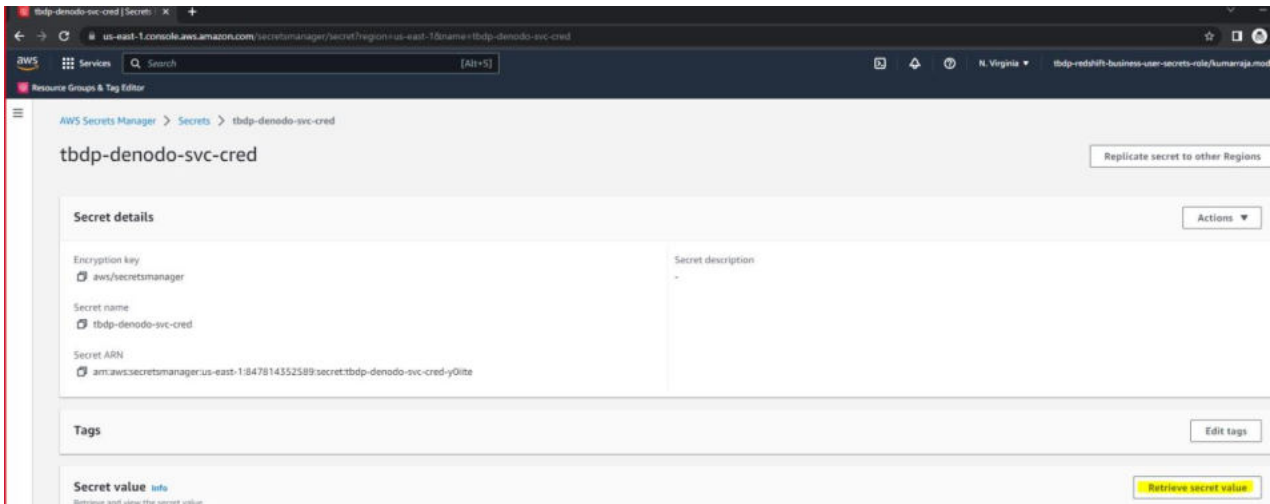
Note : Please use your TMNA login credentials

AWS URL:

<https://login.microsoftonline.com/8c642d1d-d709-47b0-ab10-080af10798fb/saml2?SAMLRequest=jZFNaa8MwDib%2FSvA9iZO6iWOSQPofhQ5G93HYTUnVNVDbneV0Y79%2BaUtPg7Gr0Cs9j1QS6ONJNYM%2FmC1%2BDEg%2B%2BNJHQxUb nFEWqCdIQCMp36mn5mGj0ogrjR524IEF60XFVllspJRNm5kJMU9mUohJPm3EUqYJXxRLF ryio96aio3hMUM04NqQB%2BPHEk95mPAwTZ%2FTRCVcJXkkc%2FF26XsEov6MFdvDkZAFDRE6P06aW0ODRveE7tx3%2BLLdVOzg%2FYIUHFP%2FbnoTwSdFoOHbmgiqOr6lspubuk r%2FbXhy1tvOjpGVdR1eD3THqMurgfvPpeBOzOol34g3csVZUbQ4RQyLLBOhmiAM292uC9s84TDd59ke2zK%2BbanL%2BPET6h8%3D&RelayState=https%3A%2F%2Fconsole.aws.amazon.com%2Fsecretsmanager%2Fhome%3Fregion%3Dus-east-1%23%21%2Fsecret%3Fname%3Dtbdbp-denodo-svc-cred>



Select Role : **tbdp-redshift-business-user-secrets-role**



Click on Retrieve Secret Value to get clientID and clientSecret.

Connection

Properties

Database Info

Data Types

Connection Properties

Database Profile

Driver Properties

Generic

Authentication

Delimited Identifiers

Qualifiers

Physical Connection

Transaction

Encoding

SQL Statements

Connection Hooks

Color and Border

SQL Commander

Query Builder

Driver Properties

Defines JDBC Driver or JNDI specific properties that can be used to fine tune the database connection. A pre-defined property is reverted to its default value when value has been edited or that the property has been added manually.

Edited	Parameter	Value
☐	connectTimeout	30000
☒	clientId	
☒	clientSecret	
☐	connectTimeout	120000
☐	debug	
☐	describeNationalCharTypesAsBasicTypes	false
☐	extraParams	
☐	i18n	
☐	identifiersUppercase	false
☐	keepAlive	
☐	loggerConsole	false
☐	loggerFile	
☐	loggerLevel	
☐	loggerNativeCalls	false
☐	nativeCompress	false
☒	oauthSSLVerify	true
☐	password	
☐	pingQuery	
☐	pingQueryTimeout	-1
☐	publishCatalogsAsSchemas	false
☐	publishViewsAsTables	false
☐	queryTimeout	900000
☐	renewTGT	
☒	scope	openid profile
☐	sendSniHeader	true
☒	ssl	true
☒	sslTrustServerCertificate	true
☐	sslTrustStoreLocation	
☐	sslTrustStorePassword	
☐	ticketCache	
☒	tokenEndpoint	https://login.microsoftonline.com/014106b1-ea3a-48aa-88f8-58cbb5ddee11/oauth2/v2.0/token
☒	useIdToken	true
☐	useKerberos	
☒	useOAuth2	true
☐	user	

☐ renewTGT

☒ scope

☐ sendSniHeader

☒ ssl

☒ sslTrustServerCertificate

☐ sslTrustStoreLocation

☐ sslTrustStorePassword

☐ ticketCache

☒ tokenEndpoint

☒ useIdToken

☐ useKerberos

☒ useOAuth2

☐ user

☐ userAgent

☐ userSSSCredential

☐ useTicketCache

☐ wanOptimized

openid

true

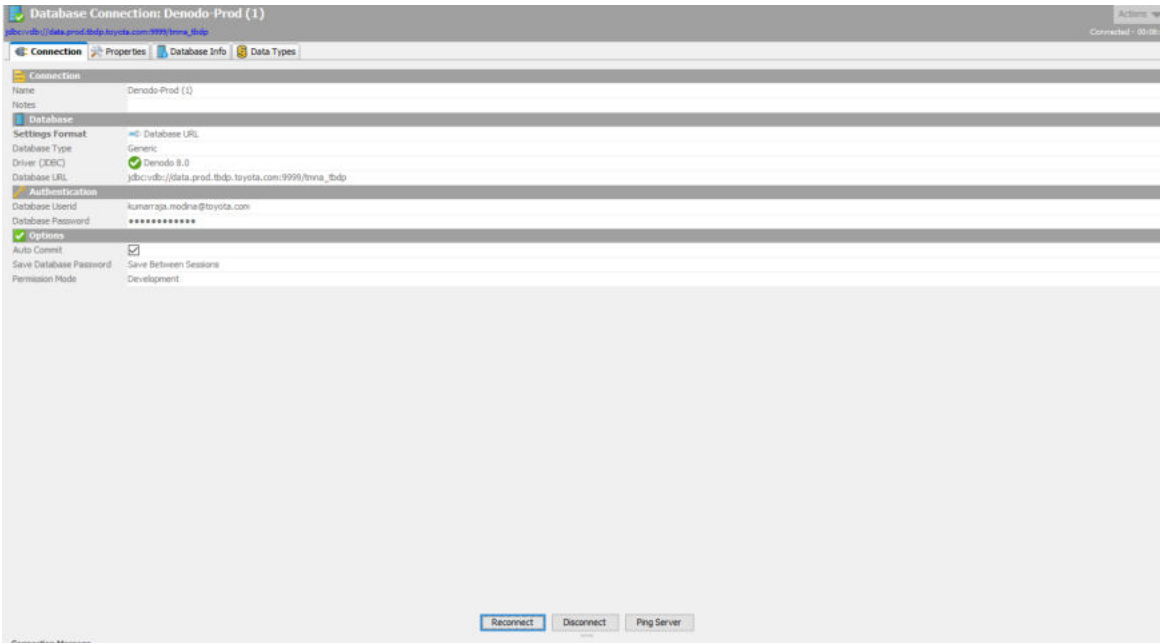
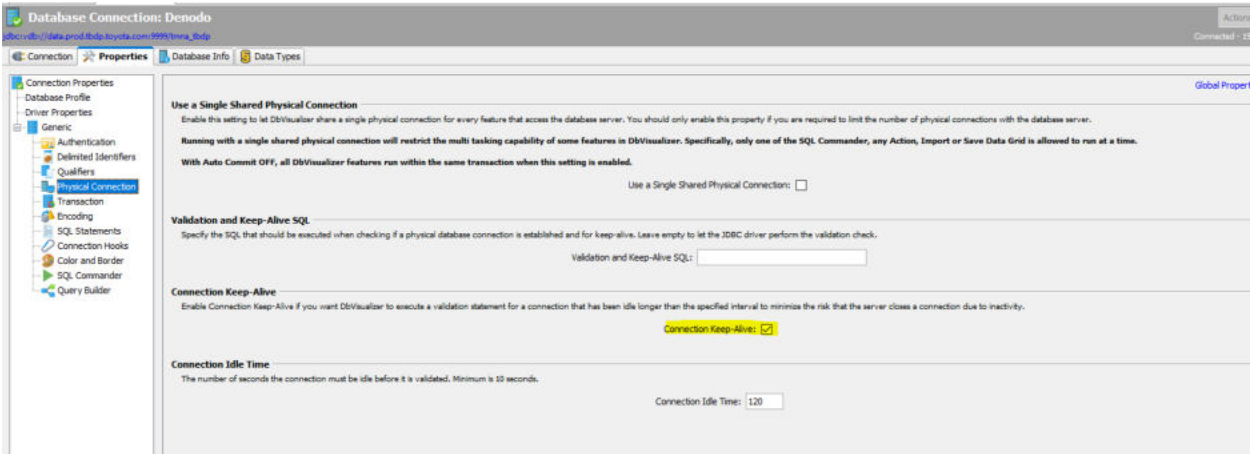
true

true

https://login.microsoftonline.com/014106b1-ea3a-48aa-88f8-58cbb5ddee11/oauth2/v2.0/token

true

true



6. Click on the Connect button, and you will establish a connection to the tmna_tbdp database. In the left panel of the window, you will see the eligible DBs and associated tables.

Microsoft SQL Server Enterprise Edition (64-bit) - SQL Server 2012

Database: DensoProd

Schema: DensoProd

Table: vehicle_logistics_frm.vehicle_accessory

Query:

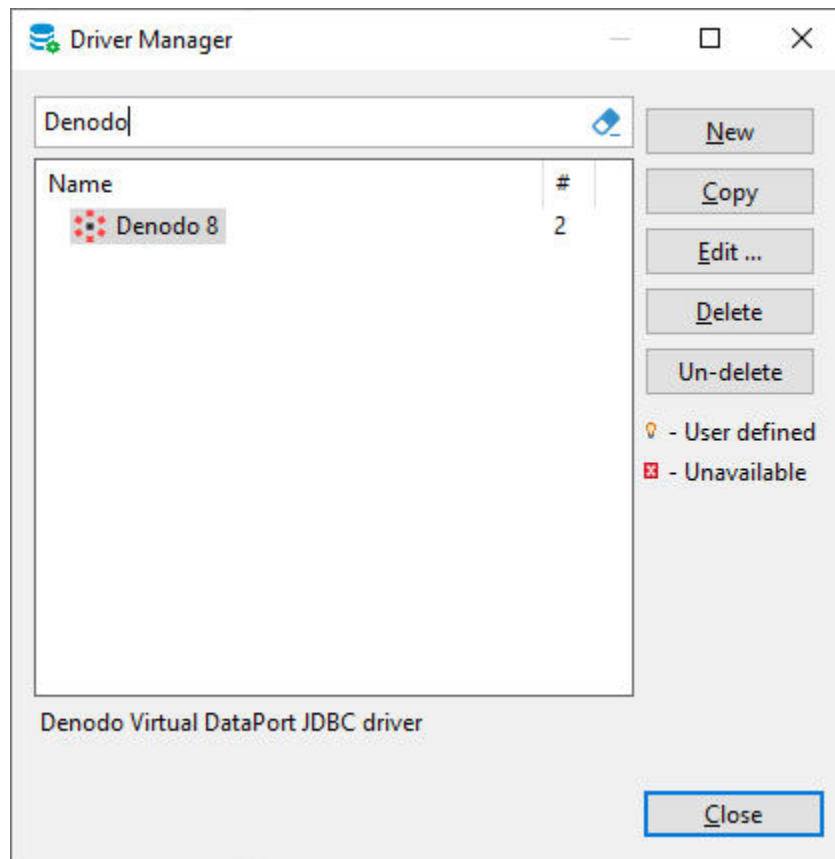
```
select * from vehicle_logistics_frm.vehicle_accessory
```

Results (1000 rows):

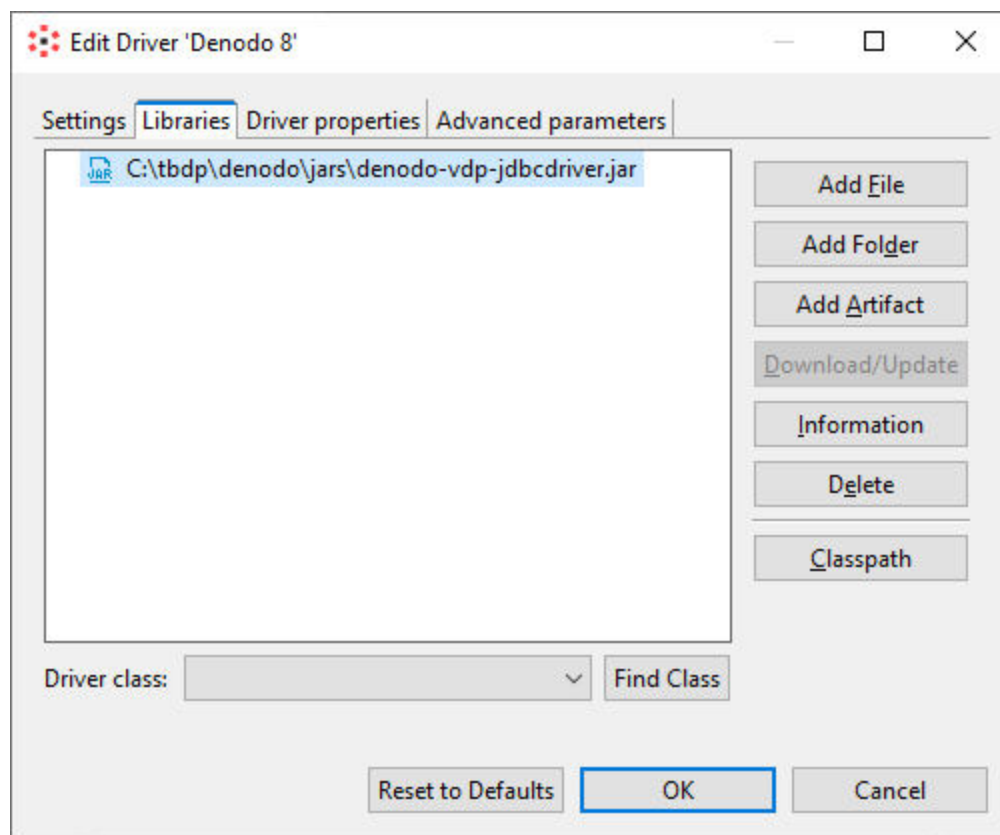
id	type	serial	model	accessory_type	accessory_code	accessory_status	price_phase	builtdet_type	psa_date	activity_date	line_sequence	line_type	dist_start_line	schedule_date	planned_period
1	FRONT	6873	05104553	6873	S	TY	1	INVT	2022-09-14 10:05:37	2022-09-02 05:14:40	U	INVT	0	INVT	
2	FRONT	6873	05104517	6873	S	TY	1	INVT	2022-09-10 01:13:39	2022-09-09 03:21:46	U	INVT	0	INVT	
3	FRONT	6873	05138718	6873	S	TY	1	INVT	2022-09-10 02:44:33	2022-09-27 03:13:58	U	INVT	0	INVT	
4	FRONT	6873	05130333	6873	S	EW	1	INVT	2022-09-14 10:30:15	2022-09-14 10:30:15	E	INVT	0	INVT	
5	FRONT	6873	05104531	6873	S	TY	1	INVT	2022-09-10 01:13:39	2022-09-20 04:19:33	U	INVT	0	INVT	
6	FRONT	6873	05130333	6873	S	TY	1	INVT	2022-09-10 01:13:39	2022-09-11 02:01:19	U	INVT	0	INVT	
7	FRONT	7531	80214888	7531	F	BY	INVT	A	2008-06-24 01:40:22	2008-06-24 01:40:22	INVT	INVT	0	INVT	
8	FRONT	8302	85465554	8302	S	CC	1	INVT	2009-01-09 10:08:42	2009-01-09 10:08:42	R	INVT	0	INVT	
9	FRONT	8302	85465483	8302	S	CC	1	INVT	2009-01-09 10:08:42	2009-01-09 10:08:42	R	INVT	0	INVT	
10	INVT	2350	81130331	2350	F	CC	INVT	A	2009-06-11 03:05:23	2009-06-11 03:05:23	INVT	INVT	0	INVT	
11	INVT	2350	81130331	2350	F	CC	INVT	A	2009-06-11 03:05:23	2009-06-11 03:05:23	INVT	INVT	0	INVT	
12	INVT	2344	81130331	2344	F	C4	1	INVT	2009-01-29 12:06:25	2009-01-29 12:06:25	U	INVT	0	INVT	
13	INVT	1934	AC940544	1934	F	EP	1	INVT	2011-03-22 08:50:37	2011-03-22 08:50:37	INVT	INVT	0	INVT	
14	INVT	2350	AC940547	2350	F	SE	1	INVT	2011-03-29 10:30:55	2011-03-29 10:30:55	R	INVT	0	INVT	
15	FRONT	7944	AD030367	7944	F	QH	1	INVT	2011-08-30 09:23:38	2011-08-30 09:23:38	U	INVT	0	INVT	
16	FRONT	1824	BF124067	1824	F	C4	P	INVT	2011-11-10 02:02:14	2011-11-10 02:02:14	U	INVT	0	INVT	

6. Accessing Denodo using a SQL client(DBeaver)

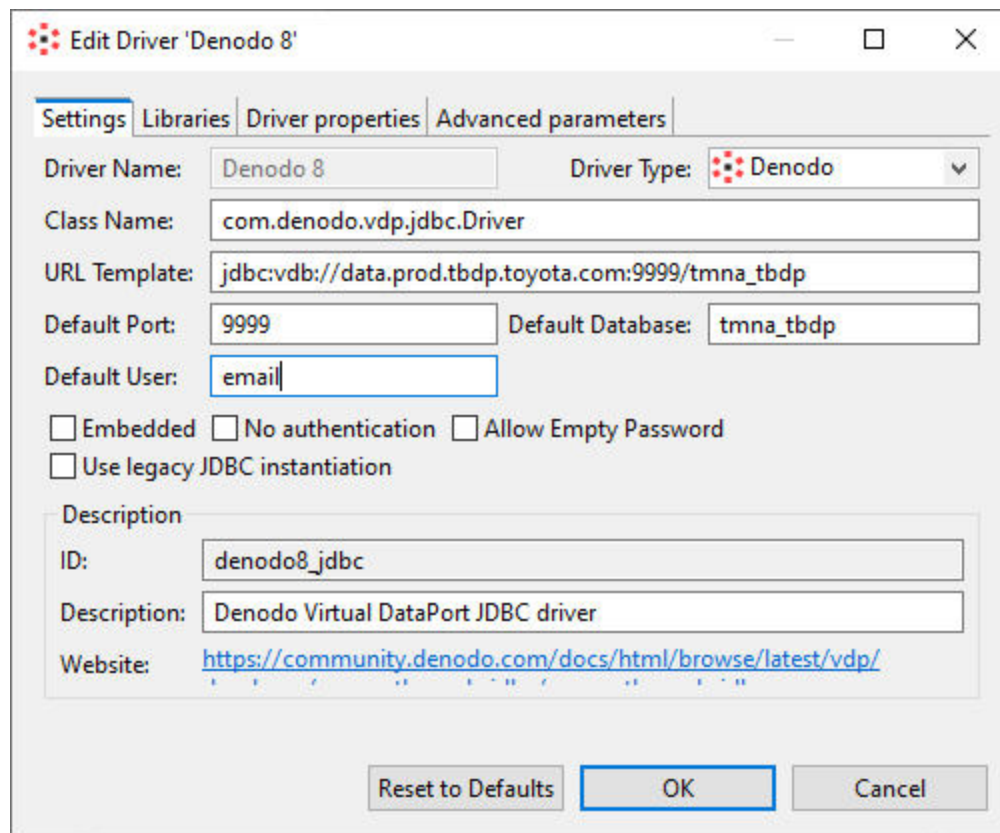
1. Add the Denodo's JDBC driver ([Denodo JDBC JAR](#)) to the client application. To use the JDBC driver in your client, you have to add the .jar file to the classpath of your application.
2. Open the DBeaver application and, in the Databases menu, select the Driver Manager option. Type the driver name Denodo and select **Denodo 8.0**



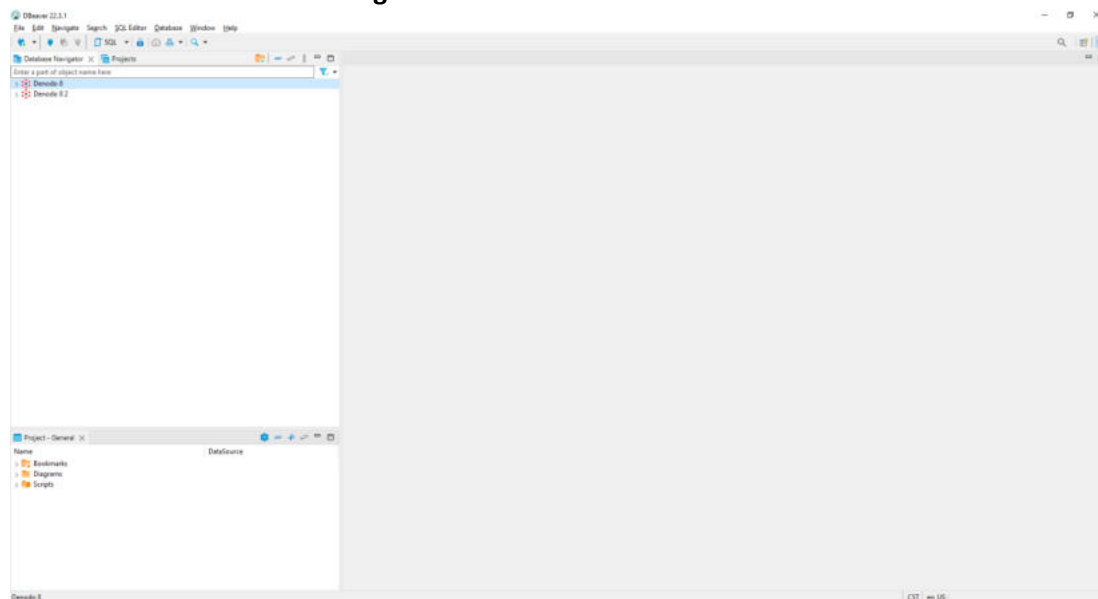
3. Double-click the **Denodo 8.0**
4. To add the .jar, click Libraries -> Add File and select the downloaded jar file



5. click Settings and provide below mentioned details.
 - a. **Class Name** : com.denodo.vdp.jdbc.Driver
 - b. **URL**: jdbc:vdb://data.prod.tbdp.toyota.com:9999/tmna_tbdp
 - c. **Default Port**: 9999
 - d. **Default Database**: tmna_tbdp
 - e. **Default User** : Provide Toyota email

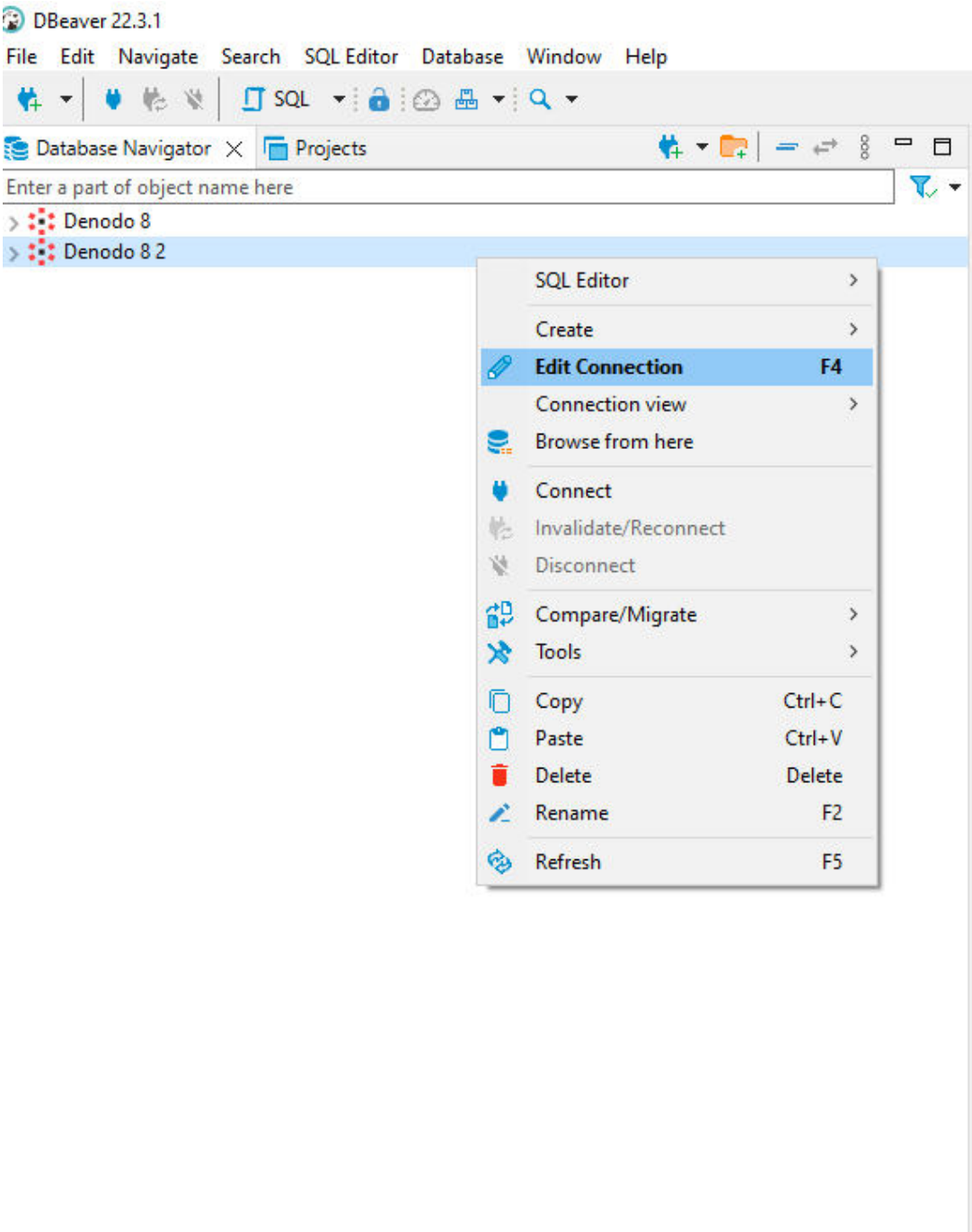


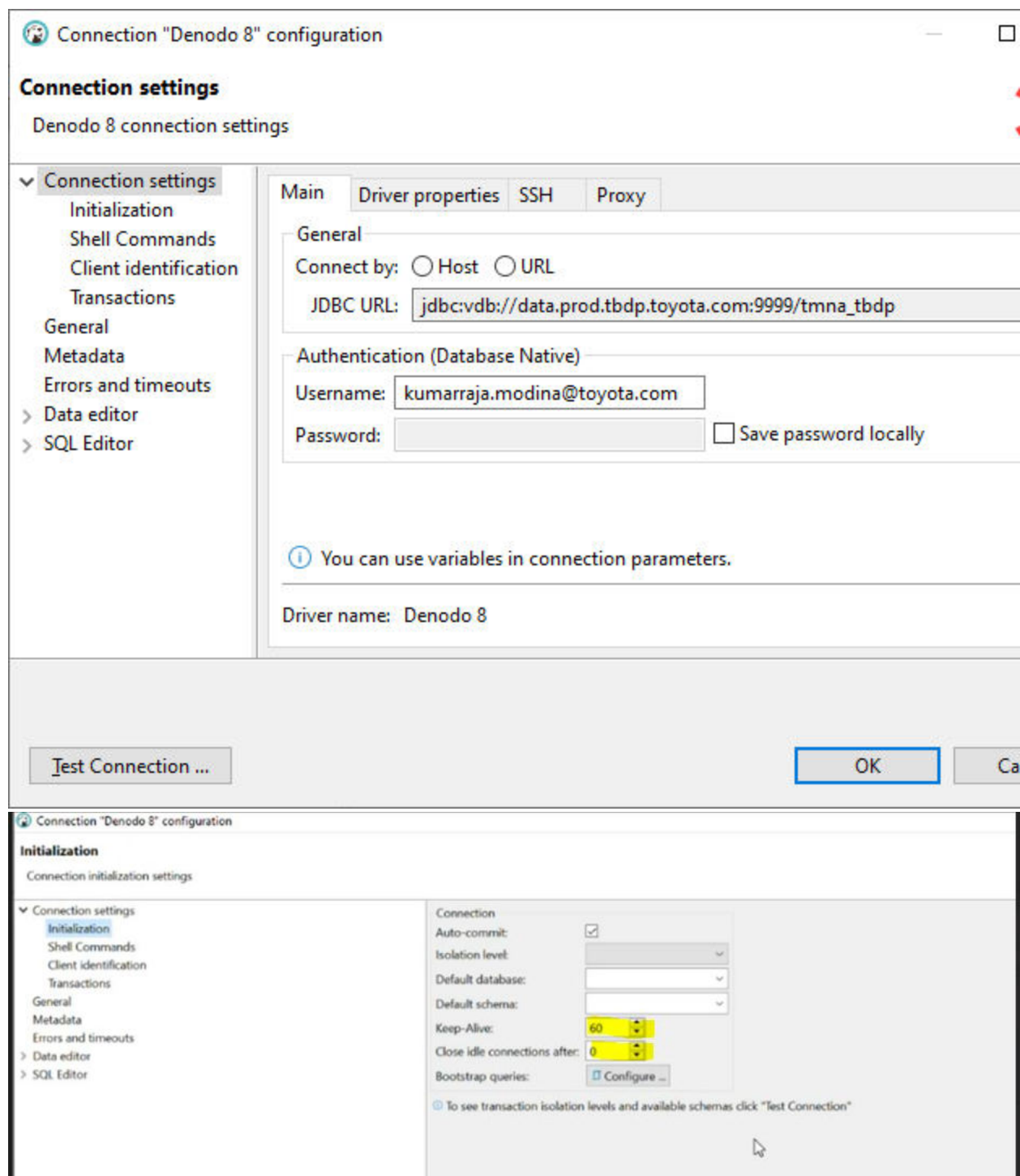
6. Click **OK** and Close **Driver Manger** window



7. You have added the driver, you can configure a connection to your Denodo virtual database.
8. Go to **Database > New Database Connection / Database Navigator** and use the following settings for the connection and Select **Denodo 8**

9. Database Navigator -> Denodo 8 -> Right click and Select Edit Connrction





10. Click on **Driver Properties**

11. Including the OAuth Parameters in Driver Properties (please refer below screenshots)

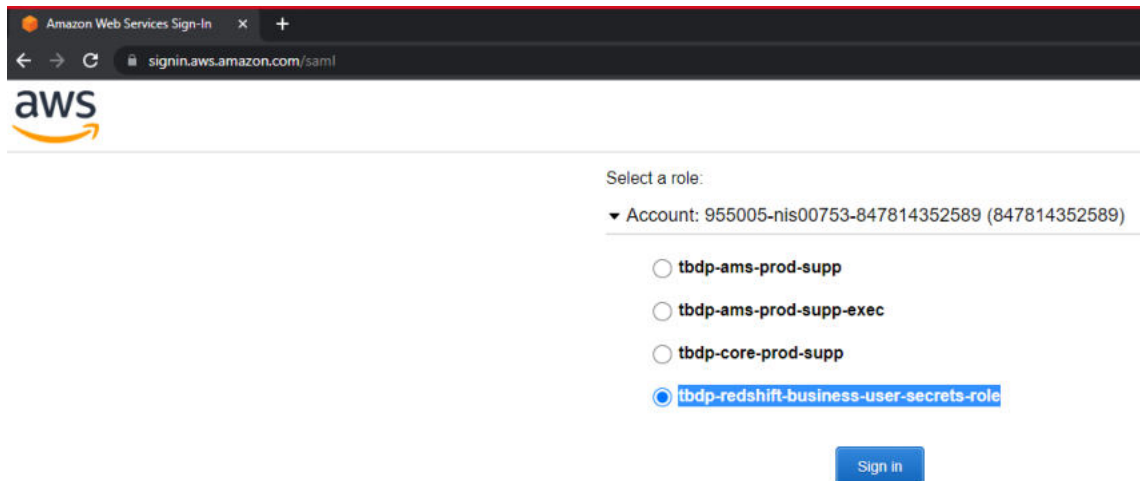
- useOAuth2: true
- TokenEndPoint: <https://login.microsoftonline.com/014106b1-ea3a-48aa-88f8-58cbb5ddee11/oauth2/v2.0/token>

- sslTrustServerCertificate: true
- scope: openid profile
- useldToken: true
- oauthSSLVerify: true
- ssl: true
- **Please retrieve the clientID and clientSecret using below AWS URL.**

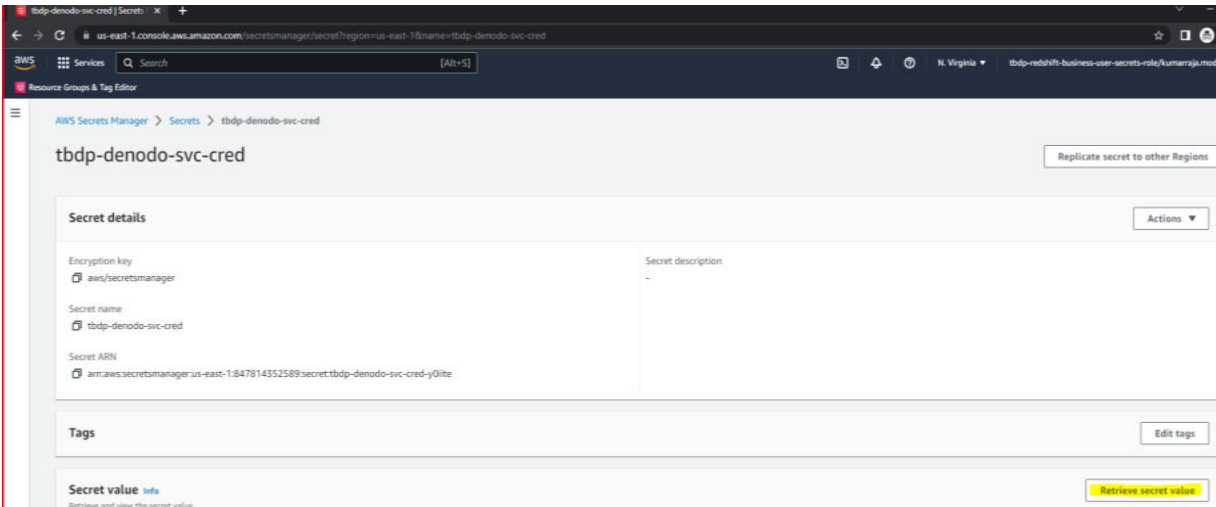
Note : Please use your TMNA login credentials

AWS URL:

<https://login.microsoftonline.com/8c642d1d-d709-47b0-ab10-080af10798fb/saml2?SAMLRequest=iZFN8MwD1b%2FSvA9iZO6iWOSQPofhQ5G93HYTUnVNVDdbneV0Y79%2BaUtPg7Gr0Cs9j1QS6ONJNYM%2FmC1%2BDEg%2B%2BNJHqXUb nFEWqCdIQCMp36mn5mGj0ogrjR524IEF60XFVIlspJRNm5kJMU9mUohJPm3EUqYJXxRLF ryio96aio3hMUM04NqQB%2BPHEk95mPAwTZ%2FTRCVcJXkkc%2FF26XsEov6MFdvDkZAFDRE6P06aW0ODRveE7tx3%2BLLdVOzg%2FYIUHFP%2FbnoTwSdFoOHbmqizOr6lspubuk r%2FbXhy1tvOjpGVdR1eD3THqMurgfvPpeBOzOoL34g3csVZUbQ4RQyLLBOhmlIAM292uC9s84TDd59ke2zK%2BbanL%2BPET6h8%3D&RelayState=https%3A%2F%2Fconsole.aws.amazon.com%2Fsecretsmanager%2Fhome%3Fregion%3Dus-east-1%23%21%2Fsecret%3Fname%3Dtbdp-denodo-svc-cred>



Select Role : **tbdp-redshift-business-user-secrets-role**



Click on Retrieve Secret Value to get clientID and clientSecret.

Connection "Denodo 8" configuration

Connection settings

Denodo 8 connection settings

▼ Connection settings

Initialization

Shell Commands

Client identification

Transactions

General

Metadata

Errors and timeouts

> Data editor

> SQL Editor

MainDriver propertiesSSHProxy

NameValue

▼ Driver properties

accessToken

adjustScaleOfDecimalValfalse

autoCommittrue

chunkSize1000

chunkTimeout90000

clientId

clientSecret

connectTimeout120000

debug

describeNationalCharTyfalse

extraParams

i18n

identifiersUppercasefalse

keepAlive

loggerConsolefalse

loggerFile

loggerLevel

loggerNativeCallsfalse

nativeCompressfalse

oauthSSLVerifytrue

pingQuery

pingQueryTimeout-1

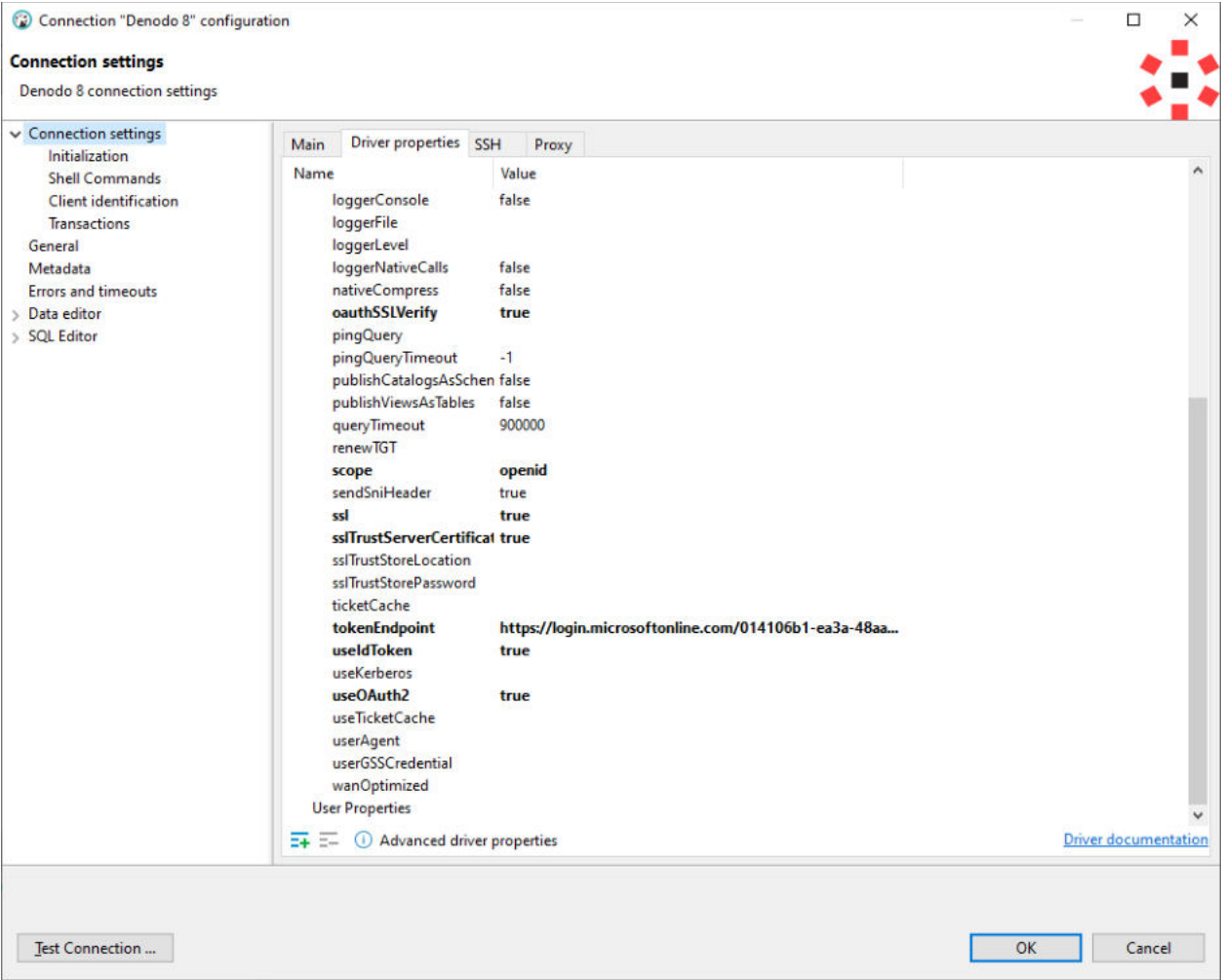
publishCatalogsAsSchenfalse

publishViewsAsTablesfalse

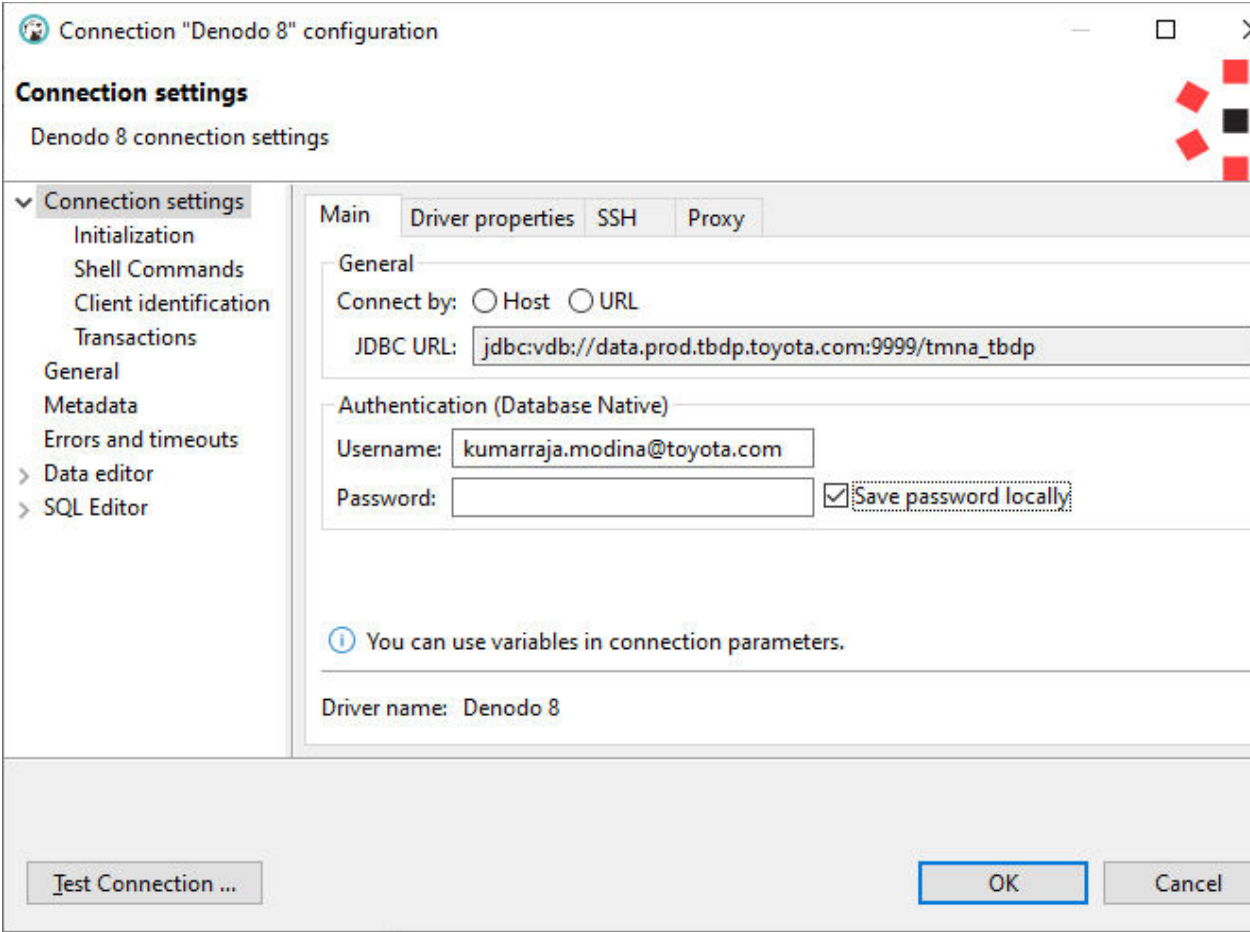
queryTimeout900000

renewTGT

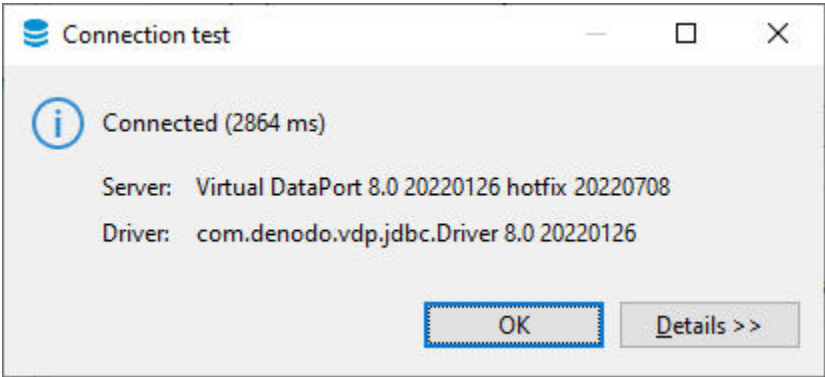
--



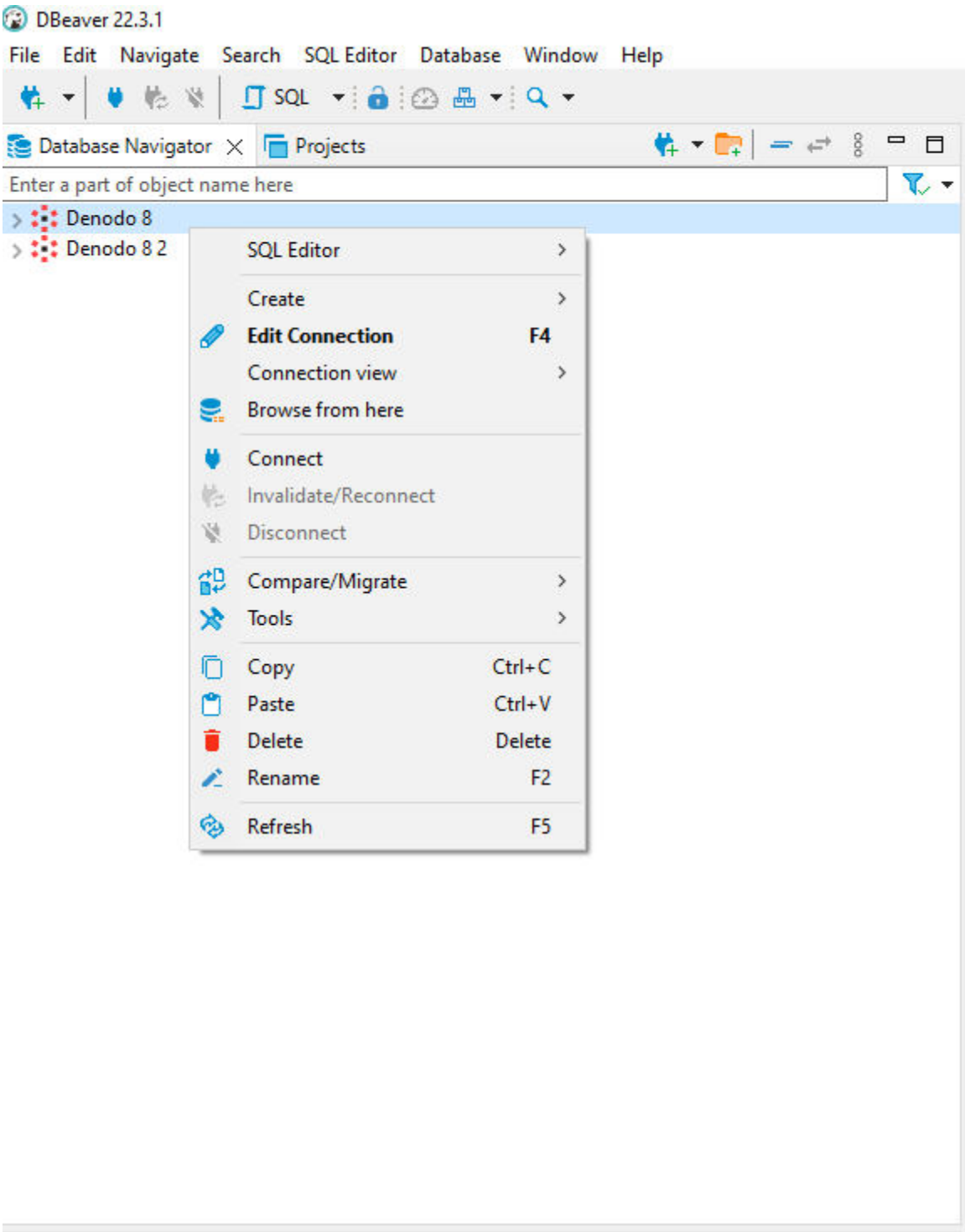
12. Enter Password: Provide Toyota network password

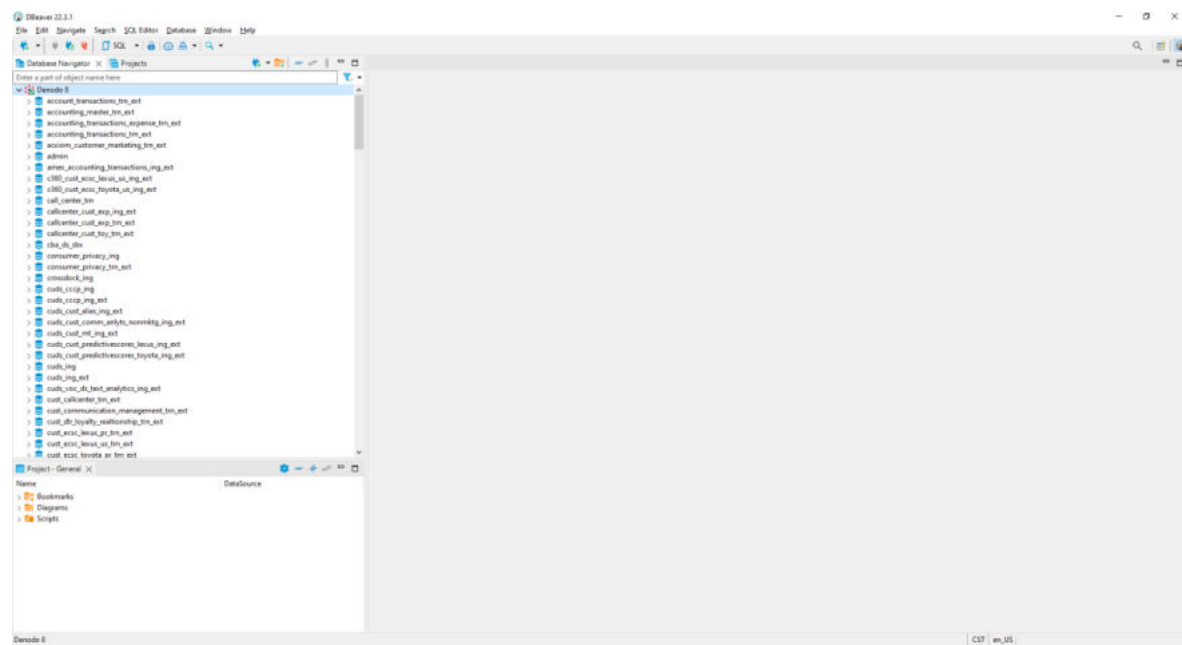


13. Click on the Test Connection and Click Ok



14. In the left panel of the window Select Denodo and Click on Connect. You will see the eligible DBs and associated tables





7. Denodo Compatible functions and Denodo reserved keywords

[Functions — VQL Guide \(denodo.com\)](#)

Below is the list of Denodo reserved keywords.

Basic elements of VQL statements - reserved words

```
<reserved words> ::=  
  ADD | ALL | ALTER | AND | ANY | AS | ASC | BASE | BOTH  
  | CASE | CONNECT | CONTEXT | CREATE | CROSS | CURRENT_DATE | CURRENT_TIMESTAMP  
  | CUSTOM | DATABASE | DEFAULT | DESC | DF | DISTINCT | DROP | EXISTS | FALSE  
  | FETCH | FLATTEN | FROM | FULL | GRANT | GROUP BY | HASH | HAVING  
  | HTML | IF | INNER | INTERSECT, INTO | IS | JDBC | JOIN | LDAP | LEADING  
  | LEFT | LIMIT | LOCALTIME | LOCALTIMESTAMP | MERGE | MINUS | MY | NATURAL | NESTED | NOS | NOT | NULL  
  | OBL | ODBC | OF | OFF | OFFSET | ON | ONE | OPT | OR | ORDER BY | ORDERED  
  | PRIVILEGES | READ | REVERSEORDER | REVOKE | RIGHT | ROW | SELECT | SWAP  
  | TABLE | TO | TRACE | TRAILING | TRUE | UNION | USER | USING | VIEW | WHEN  
  | WHERE | WITH | WRITE | WS | ZERO
```

8. Best Practices for migrating SQL Queries to Denodo VQL

<https://community.denodo.com/kb/en/view/document/Best%20Practices%20for%20Migrating%20SQL%20queries%20to%20Denodo%20VQL>